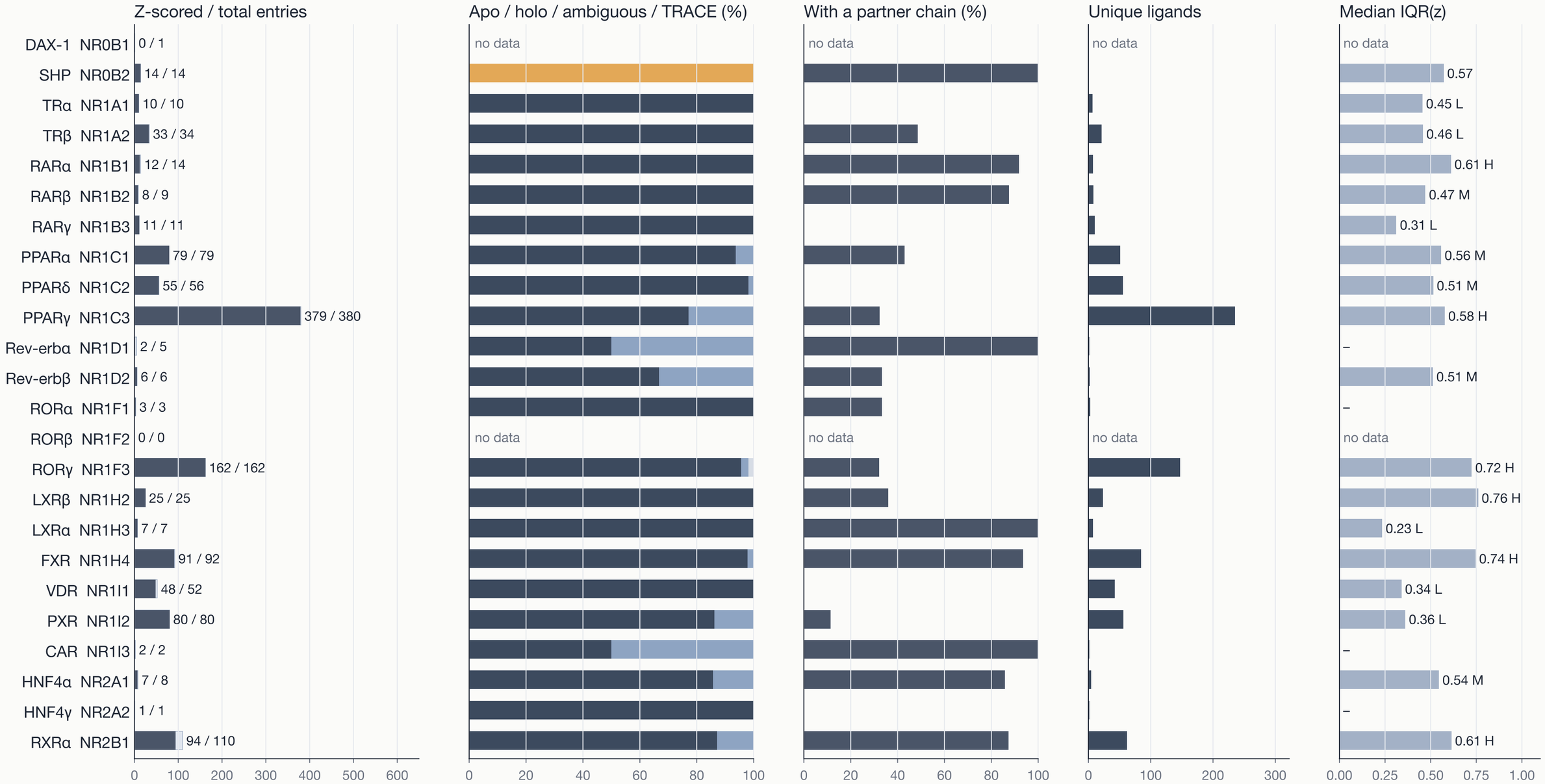
Dataset at a glance (1/2)

One row per receptor. First panel: entries with z-scores (dark) of all PDB entries found (light). Other panels use the z-scored entries.



Apo/holo colours: HOLO dark, APO mid, AMBIGUOUS light, TRACE_LBD orange. Heterogeneity letter: L/M/H = low/moderate/high tertile.

Dataset at a glance (2/2)

One row per receptor. First panel: entries with z-scores (dark) of all PDB entries found (light). Other panels use the z-scored entries.



Apo/holo colours: HOLO dark, APO mid, AMBIGUOUS light, TRACE_LBD orange. Heterogeneity letter: L/M/H = low/moderate/high tertile.

DAX-1 NR0B1 · 0 usable PDBs

no usable LBD structures

STRUCTURES

No usable LBD structures

1 entry found (receptor chain outside the LBD (1))

LIGANDS

–

PARTNERS / COFACTORS

–

no professor helix map



BETWEEN-PDB HETEROGENEITY

not assessable

No structures

CAVEATS

- No professor helix map: no canonical elements.

SHP NR0B2 · 14 usable PDBs

fragment-only · no bound ligands · partner-rich · IQR not assessable

STRUCTURES

TRACE 14

14 found · 14 with coordinates · 5% of LBD observed (median)

LIGANDS

no bound ligands

PARTNERS / COFACTORS

NR partner 13

other 1

most common: ancestral glucocorticoid receptor2

no professor helix map



BETWEEN-PDB HETEROGENEITY

not assessable

Only 13 positions in ≥5 structures

CONSTRUCTS & CRYSTALS

8 construct patterns · 12 crystal forms

2.35 Å median (1.95–2.58)

CAVEATS

- No professor helix map: no canonical elements.
- 14 TRACE_LBD fragments (receptor peptides) excluded from apo/holo.
- Very heterogeneous crystal forms (12 forms in 14 structures).

TRα NR1A1 · 10 usable PDBs

holo only · one dominant ligand · mostly standalone · low IQR heterogeneity

STRUCTURES

HOLO 10 no APO

10 found · 10 with coordinates · 100% of LBD observed (median)

LIGANDS

6 unique · T3 in 50%

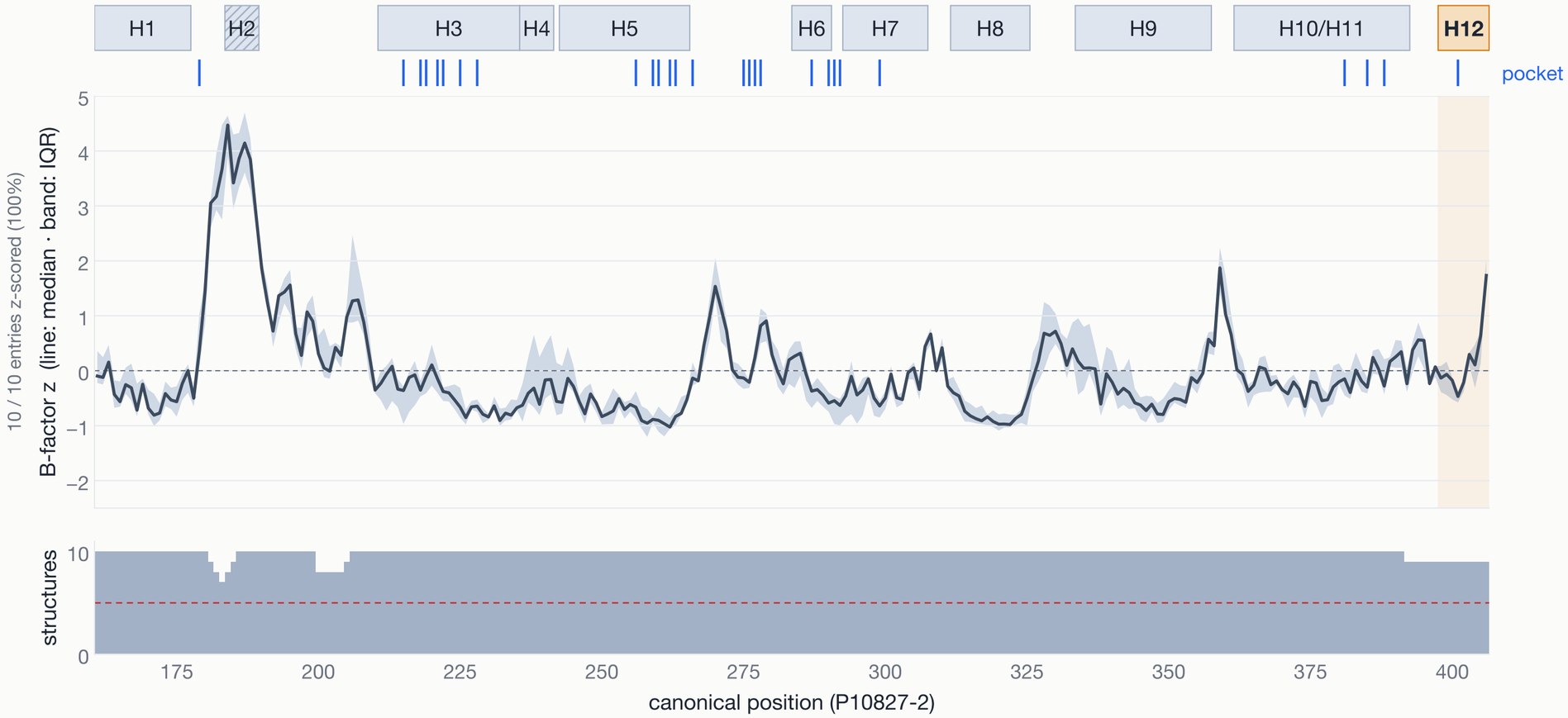
T3 5 · B72 2 · IH5 1

6 ligand–partner contexts

dominant: 3,5,3'triiodothyronine + standalone (40%)

PARTNERS / COFACTORS

Standalone: 100%



BETWEEN-PDB HETEROGENEITY

LOW

median IQR	0.45
90th percentile	0.85
supported positions	246 / 246

CONSTRUCTS & CRYSTALS

3 construct patterns · 5 crystal forms

2.19 Å median (1.93–2.50)

CAVEATS

- Very heterogeneous crystal forms (5 forms in 10 structures).

TRβ NR1A2 · 33 usable PDBs

holo only · ligand-diverse · partly partner-bound · low IQR heterogeneity

STRUCTURES

HOLO 33 — no APO

34 found · 33 with coordinates · 95% of LBD observed (median)

LIGANDS

21 unique

4HY 7 · 8HO 5 · T3 4

23 ligand–partner contexts

dominant: 4HY + standalone (18%)

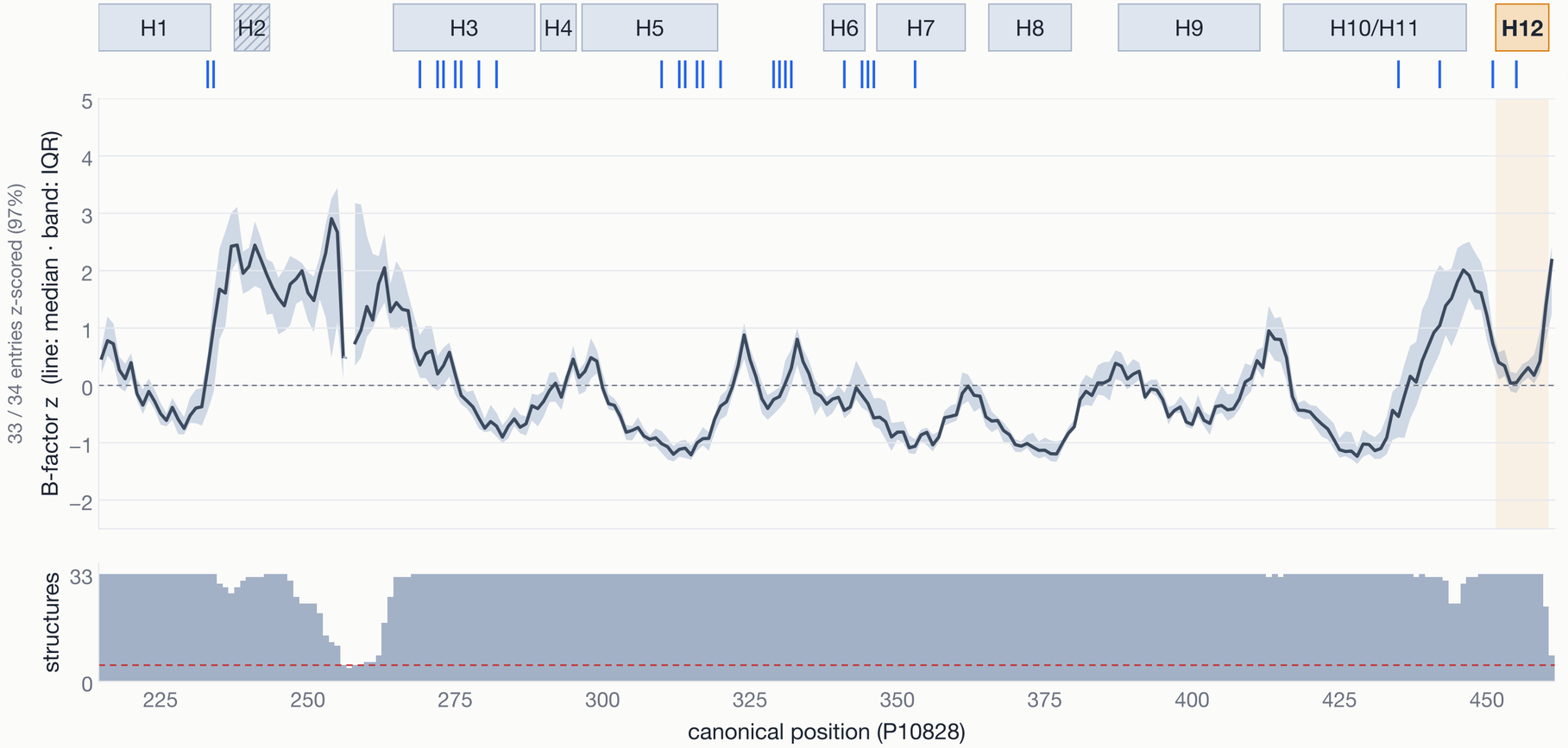
PARTNERS / COFACTORS

coregulator 15 | standalone 17

NR partner 1

most common: nuclear receptor coactivator 2

2 unique partner identities



BETWEEN-PDB HETEROGENEITY

LOW

median IQR	0.46
90th percentile	1.12
supported positions	246 / 247

CONSTRUCTS & CRYSTALS

9 construct patterns · 11 crystal forms

2.65 Å median (2.40–2.81)

engineered mutants: 27% (top D355R)

RARα NR1B1 · 12 usable PDBs

holo only · mixed ligands · partner-rich · high IQR heterogeneity

STRUCTURES

HOLO 12 — no APO

14 found · 12 with coordinates · 99% of LBD observed (median)

LIGANDS

7 unique

EQN 5 · BMS 2 · REA 1

10 ligand–partner contexts

dominant: BMS + RXRα (17%)

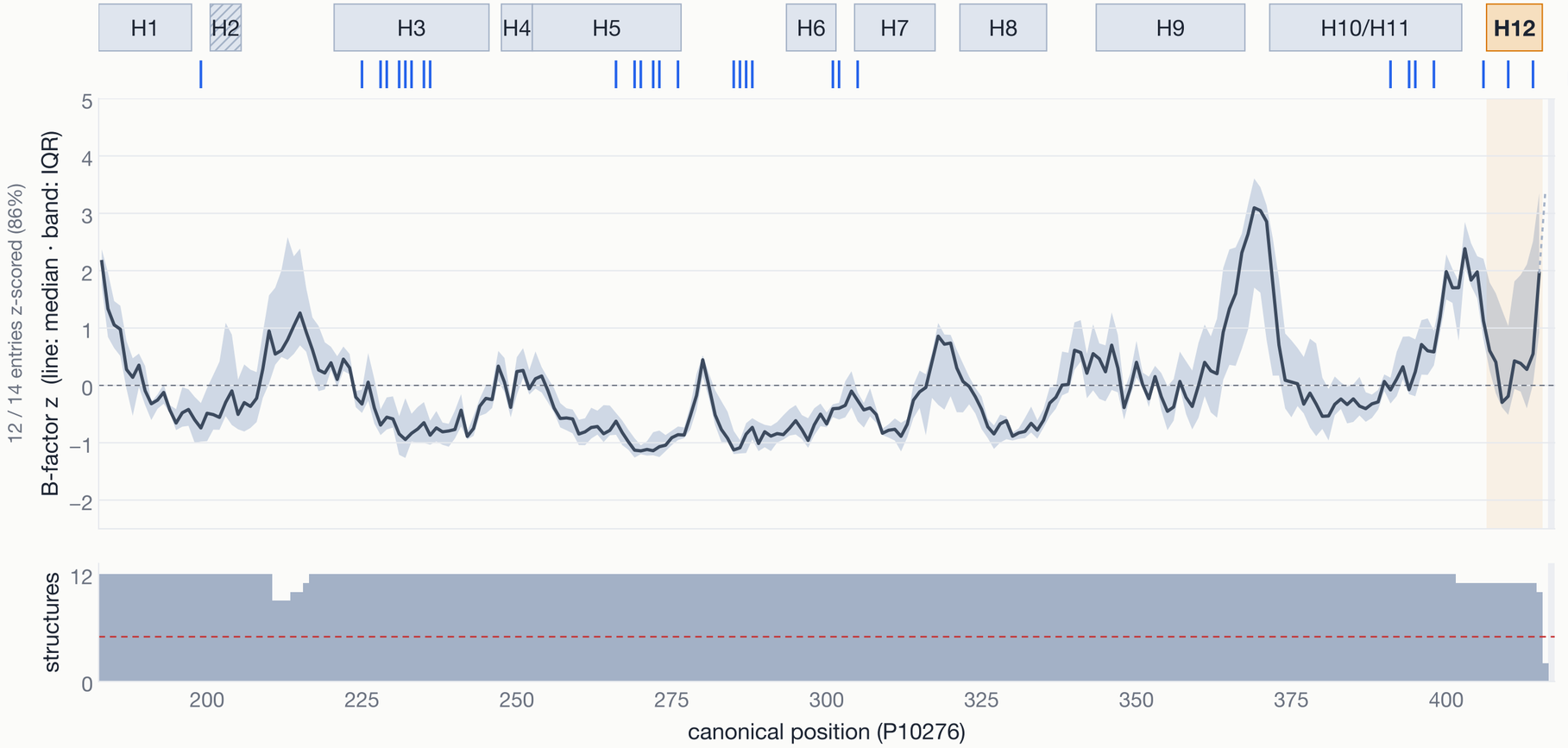
PARTNERS / COFACTORS

coregulator 5 | NR partner 4 | other 2

standalone 1

most common: retinoic acid receptor rxr-alpha

5 unique partner identities



BETWEEN-PDB HETEROGENEITY

HIGH

median IQR	0.61
90th percentile	1.57
supported positions	233 / 235

CONSTRUCTS & CRYSTALS

7 construct patterns · 9 crystal forms

2.25 Å median (1.89–2.60)

CAVEATS

- Very heterogeneous crystal forms (9 forms in 12 structures).

RARβ NR1B2 · 8 usable PDBs

holo only · ligand-diverse · partner-rich · moderate IQR heterogeneity

STRUCTURES

HOLO 8 — no APO

9 found · 8 with coordinates · 99% of LBD observed (median)

LIGANDS

8 unique

TTB 2 · REA 2 · 1NY 1

8 ligand–partner contexts

dominant: TTB + standalone (12%)

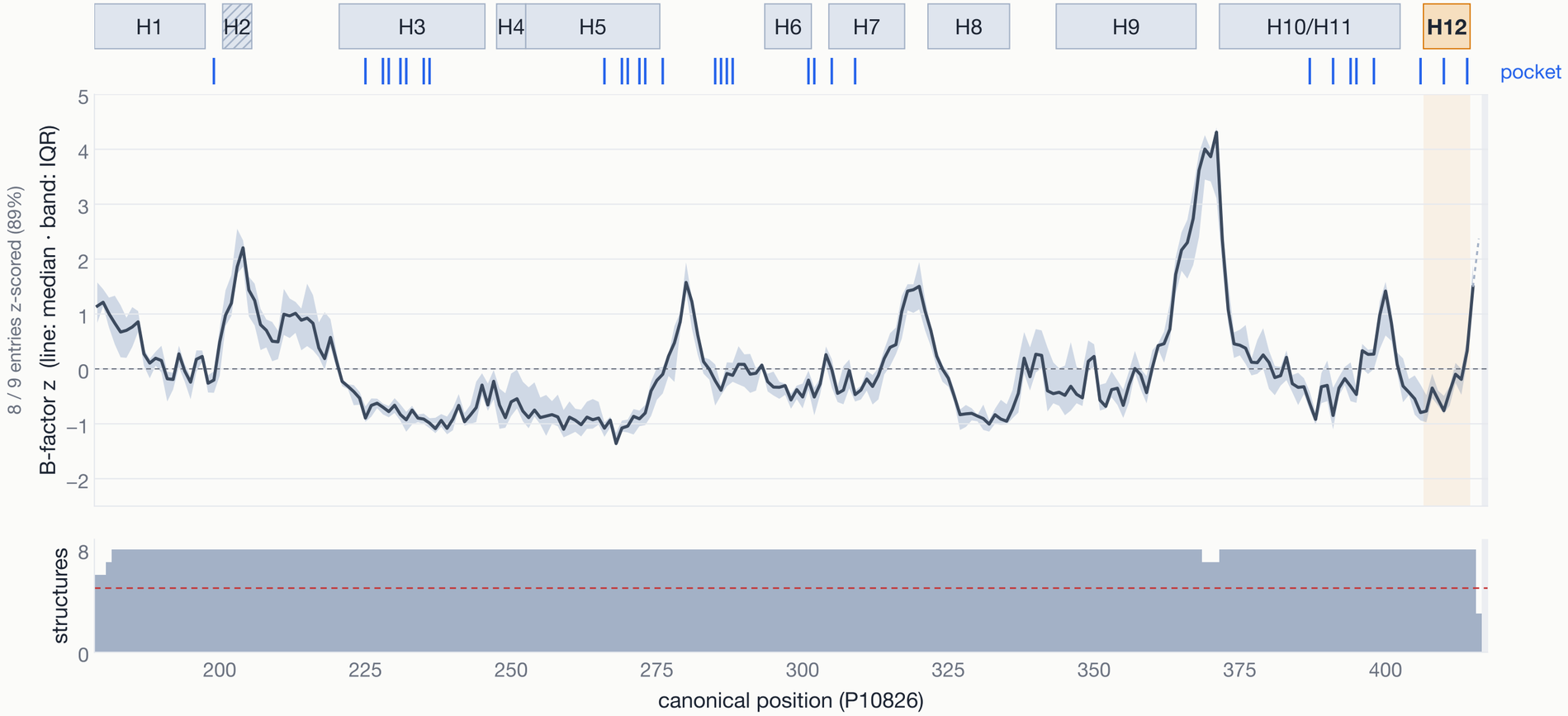
PARTNERS / COFACTORS

coregulator 6

NR partner 1 standalone 1

most common: nuclear receptor coactivator 1

3 unique partner identities



BETWEEN-PDB HETEROGENEITY

MODERATE

median IQR	0.47
90th percentile	0.78
supported positions	237 / 239

CONSTRUCTS & CRYSTALS

2 construct patterns · 5 crystal forms

2.30 Å median (2.05–2.41)

substitution status uncertain: 50%

RARγ NR1B3 · 11 usable PDBs

holo only · one dominant ligand · mostly standalone · low IQR heterogeneity

STRUCTURES

HOLO 11 — no APO

11 found · 11 with coordinates · 99% of LBD observed (median)

LIGANDS

10 unique · LMU in 64%

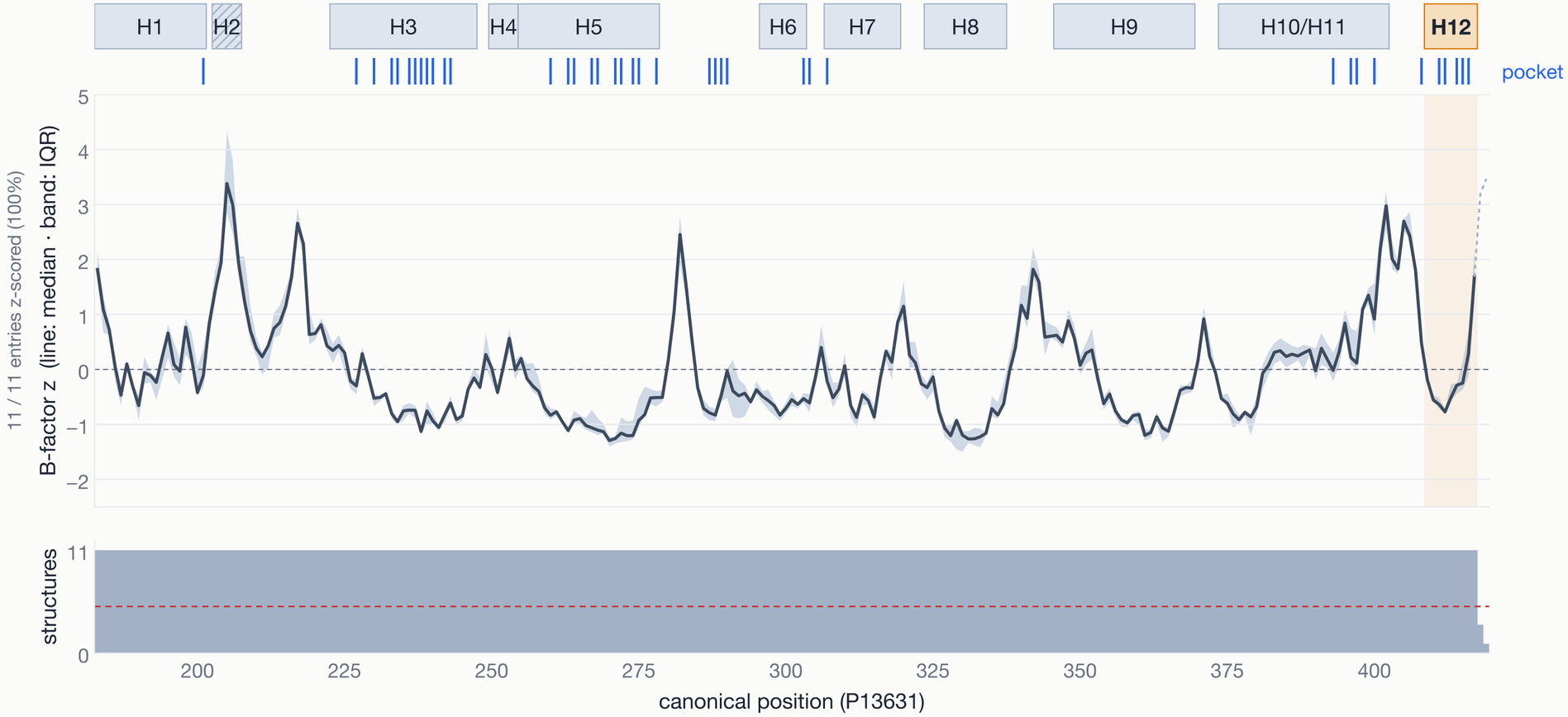
LMU 7 · 961 2 · 9CR 2

11 ligand–partner contexts

dominant: 394 + LMU + standalone (9%)

PARTNERS / COFACTORS

Standalone: 100%



BETWEEN-PDB HETEROGENEITY

LOW

median IQR	0.31
90th percentile	0.58
supported positions	235 / 237

CONSTRUCTS & CRYSTALS

3 construct patterns · 1 crystal form

1.67 Å median (1.42–1.98)

PPARα NR1C1 · 79 usable PDBs

mostly holo · ligand-diverse · partly partner-bound · moderate IQR heterogeneity

STRUCTURES

HOLO 74

APO 5

79 found · 79 with coordinates · 98% of LBD observed (median)

LIGANDS

51 unique

2VN 6 · PLM 5 · P7F 5

61 ligand–partner contexts

dominant: WY1 + standalone (5%)

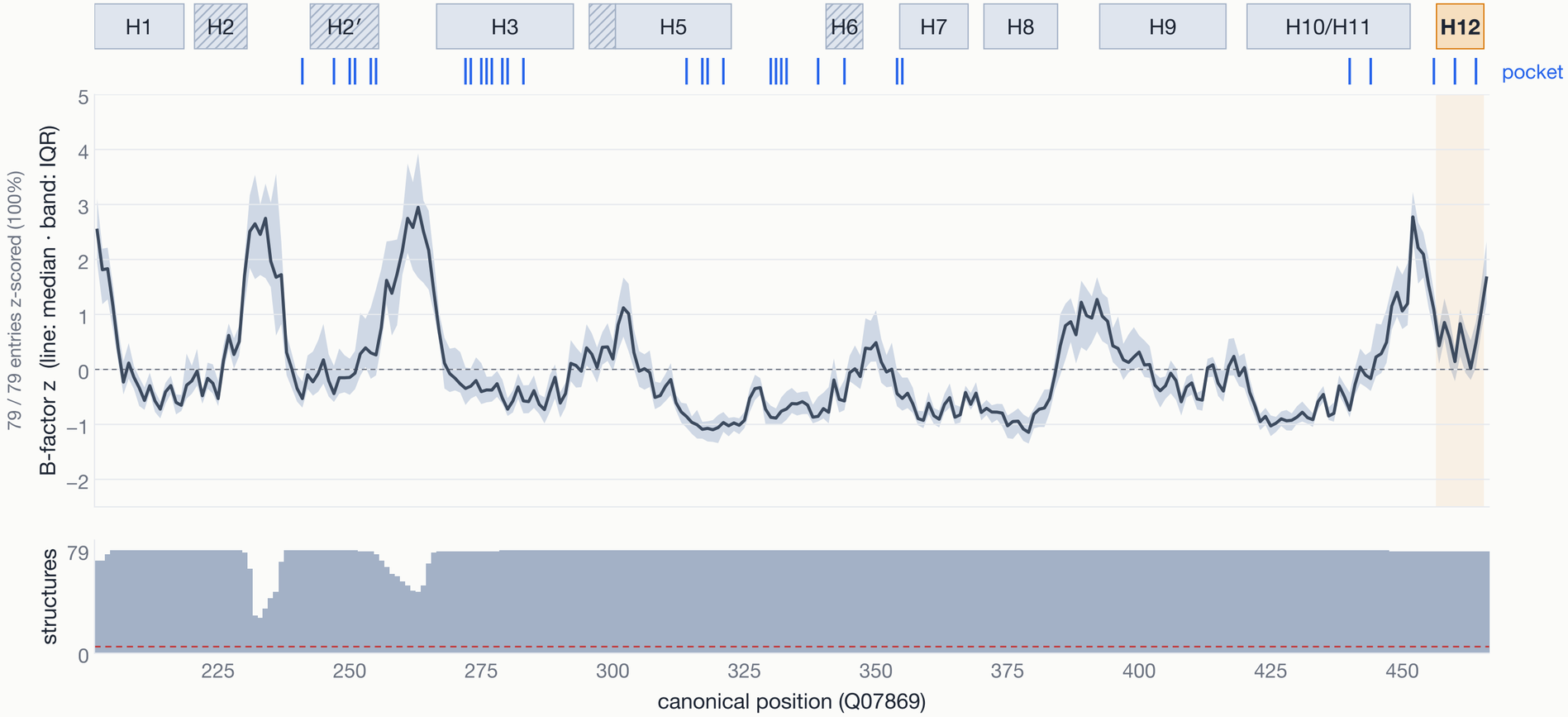
PARTNERS / COFACTORS

coregulator 31 | standalone 45

other 3

most common: peroxisome proliferator-activated…

7 unique partner identities



BETWEEN-PDB HETEROGENEITY

MODERATE

median IQR	0.56
90th percentile	1.10
supported positions	265 / 265

CONSTRUCTS & CRYSTALS

9 construct patterns · 24 crystal forms

1.95 Å median (1.56–2.30)

PPARδ NR1C2 · 55 usable PDBs

mostly holo · ligand-diverse · mostly standalone · moderate IQR heterogeneity

STRUCTURES

HOLO 54

APO 1

56 found · 55 with coordinates · 98% of LBD observed (median)

LIGANDS

55 unique

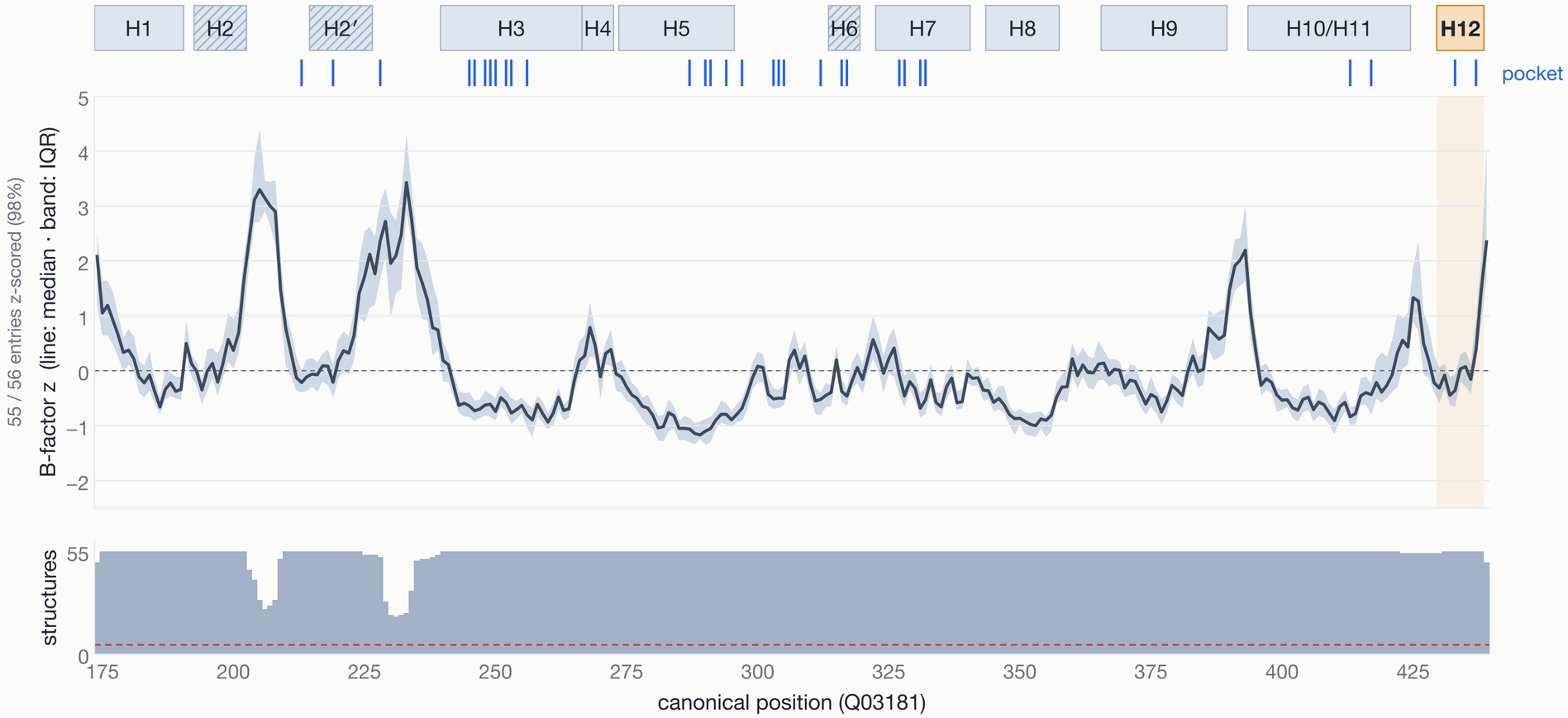
VCA 3 · B7G 3 · K55 2

53 ligand–partner contexts

dominant: vaccenic acid + standalone (6%)

PARTNERS / COFACTORS

Standalone: 100%



BETWEEN-PDB HETEROGENEITY

MODERATE

median IQR	0.51
90th percentile	1.01
supported positions	266 / 266

CONSTRUCTS & CRYSTALS

10 construct patterns · 12 crystal forms

2.10 Å median (1.95–2.45)

PPARγ NR1C3 · 379 usable PDBs

mixed apo/holo · ligand-diverse · partly partner-bound · high IQR heterogeneity

STRUCTURES

HOLO 292 | APO 87

380 found · 379 with coordinates · 96% of LBD observed (median)

LIGANDS

235 unique

BRL 14 · EDK 13 · KNA 8

265 ligand–partner contexts

dominant: APO + standalone (13%)

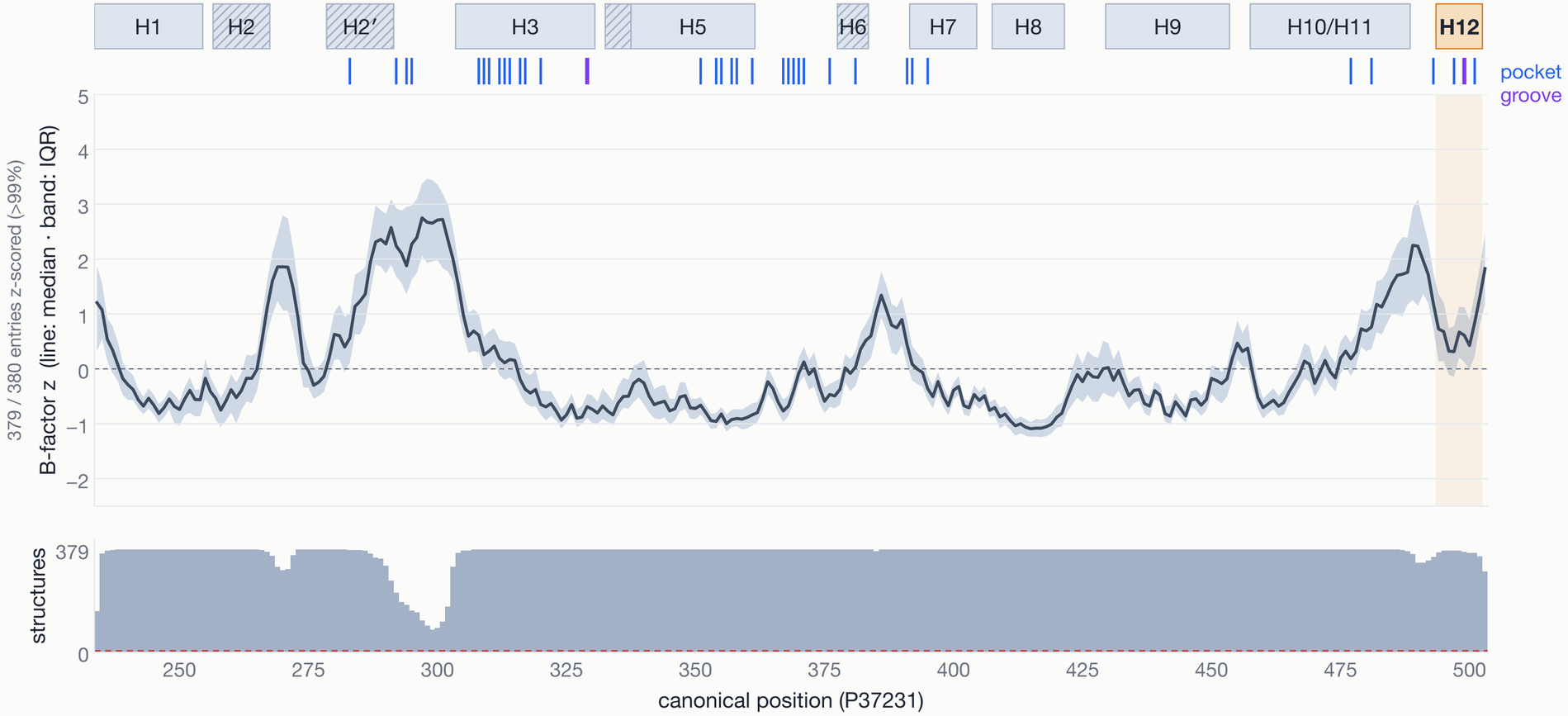
PARTNERS / COFACTORS

coregulator 99 | standalone 257

NR partner 9 other 14

most common: nuclear receptor coactivator 1

16 unique partner identities



BETWEEN-PDB HETEROGENEITY

HIGH

median IQR	0.58
90th percentile	1.14
supported positions	270 / 270

CONSTRUCTS & CRYSTALS

18 construct patterns · 79 crystal forms

2.25 Å median (2.07–2.48)

Rev-erbα NR1D1 · 2 usable PDBs

mixed apo/holo · one dominant ligand · partner-rich · IQR not assessable

STRUCTURES

HOLO 1 | APO 1

5 found · 2 with coordinates · 58% of LBD observed (median)

LIGANDS

1 unique · QFX in 100%

QFX 1

2 ligand–partner contexts

dominant: APO + NCOR1 (50%)

PARTNERS / COFACTORS

coregulator 2

most common: nuclear receptor corepressor 1

1 unique partner identity



BETWEEN-PDB HETEROGENEITY

not assessable

Only 2 usable structures

CONSTRUCTS & CRYSTALS

1 construct pattern · 2 crystal forms

2.55 Å median

2 internal deletion

CAVEATS

- 2 of 2 constructs carry an internal deletion.

Rev-erbβ NR1D2 · 6 usable PDBs

mixed apo/holo · one dominant ligand · partly partner-bound · moderate IQR heterogeneity

STRUCTURES

HOLO 4 | APO 2

6 found · 6 with coordinates · 93% of LBD observed (median)

LIGANDS

2 unique · HEM in 75%

HEM 3 · COH 1

5 ligand–partner contexts

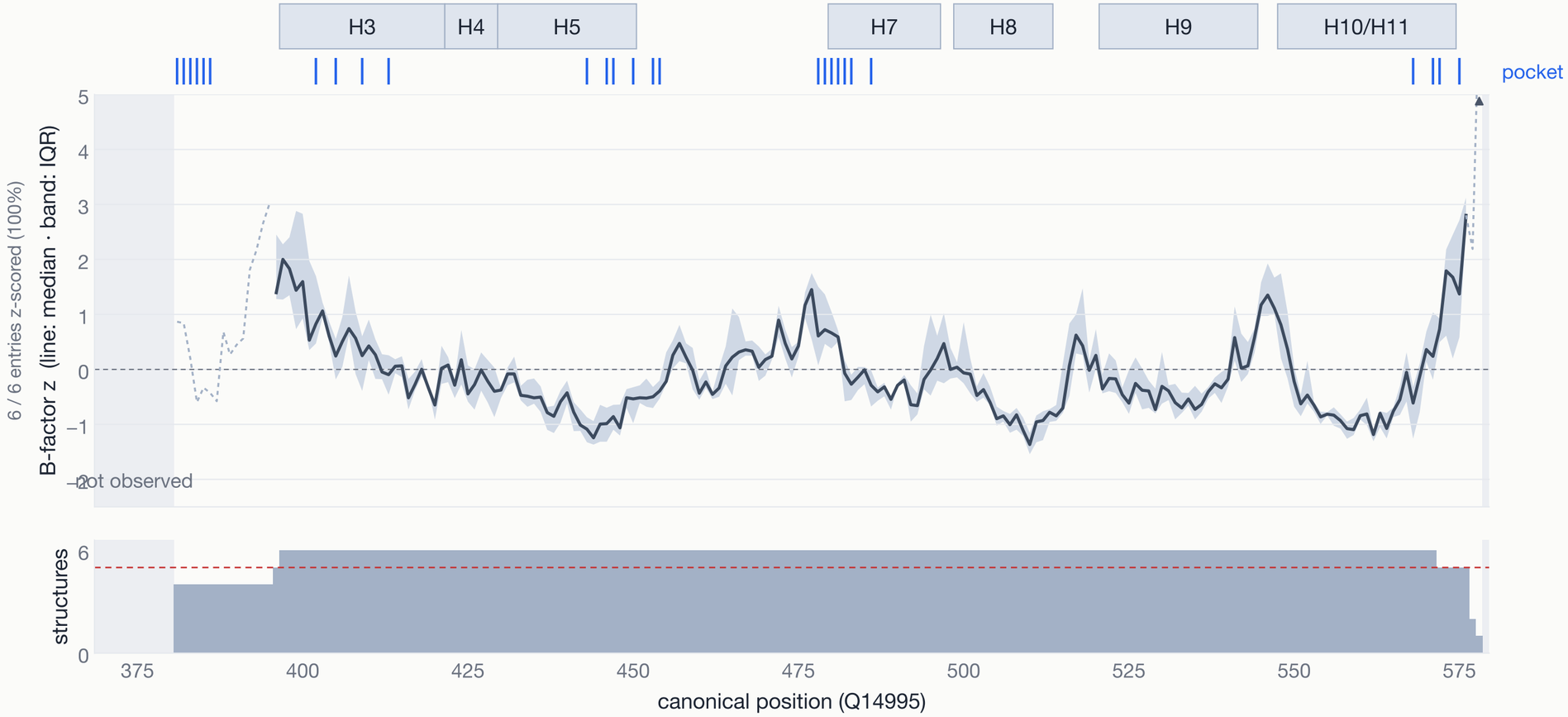
dominant: APO + standalone (33%)

PARTNERS / COFACTORS

coregulator 2 | standalone 4

most common: nuclear receptor corepressor 1

2 unique partner identities



BETWEEN-PDB HETEROGENEITY

MODERATE

median IQR	0.51
90th percentile	1.14
supported positions	181 / 211

CONSTRUCTS & CRYSTALS

3 construct patterns · 5 crystal forms

2.20 Å median (1.92–2.40)

substitution status uncertain: 33%

RORα NR1F1 · 3 usable PDBs

holo only · ligand-diverse · partly partner-bound · IQR not assessable

STRUCTURES

HOLO 3 — no APO

3 found · 3 with coordinates · 100% of LBD observed (median)

LIGANDS

3 unique

CLR 1 · C3S 1 · 4D8 1

3 ligand–partner contexts

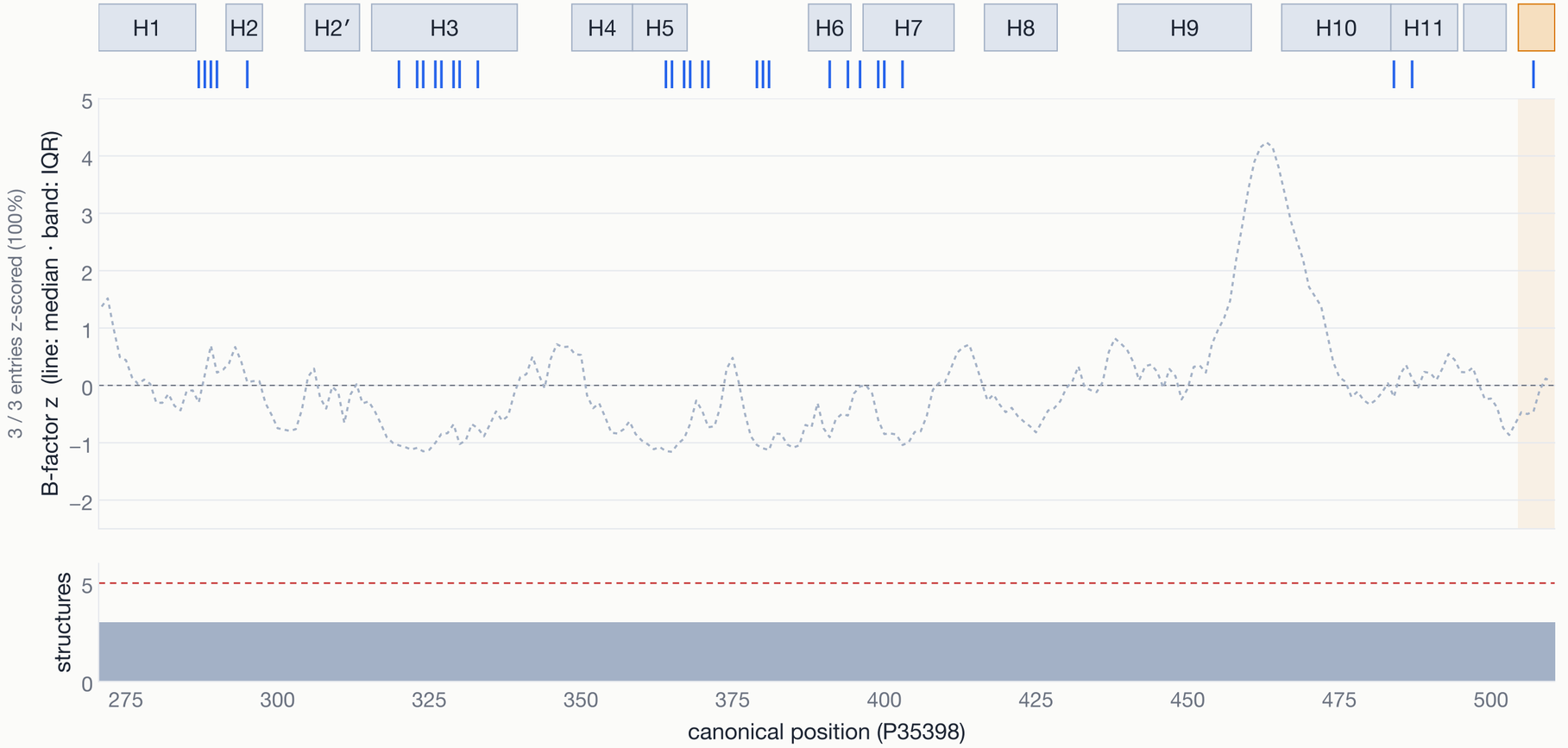
dominant: cholesterol + standalone (33%)

PARTNERS / COFACTORS

other 1 | standalone 2

most common: nuclear receptor-interacting protein 1

1 unique partner identity



BETWEEN-PDB HETEROGENEITY

not assessable

Only 3 usable structures

CONSTRUCTS & CRYSTALS

2 construct patterns · 2 crystal forms

1.90 Å median

RORβ NR1F2 · 0 usable PDBs

no usable LBD structures

STRUCTURES

No usable LBD structures

0 entries found

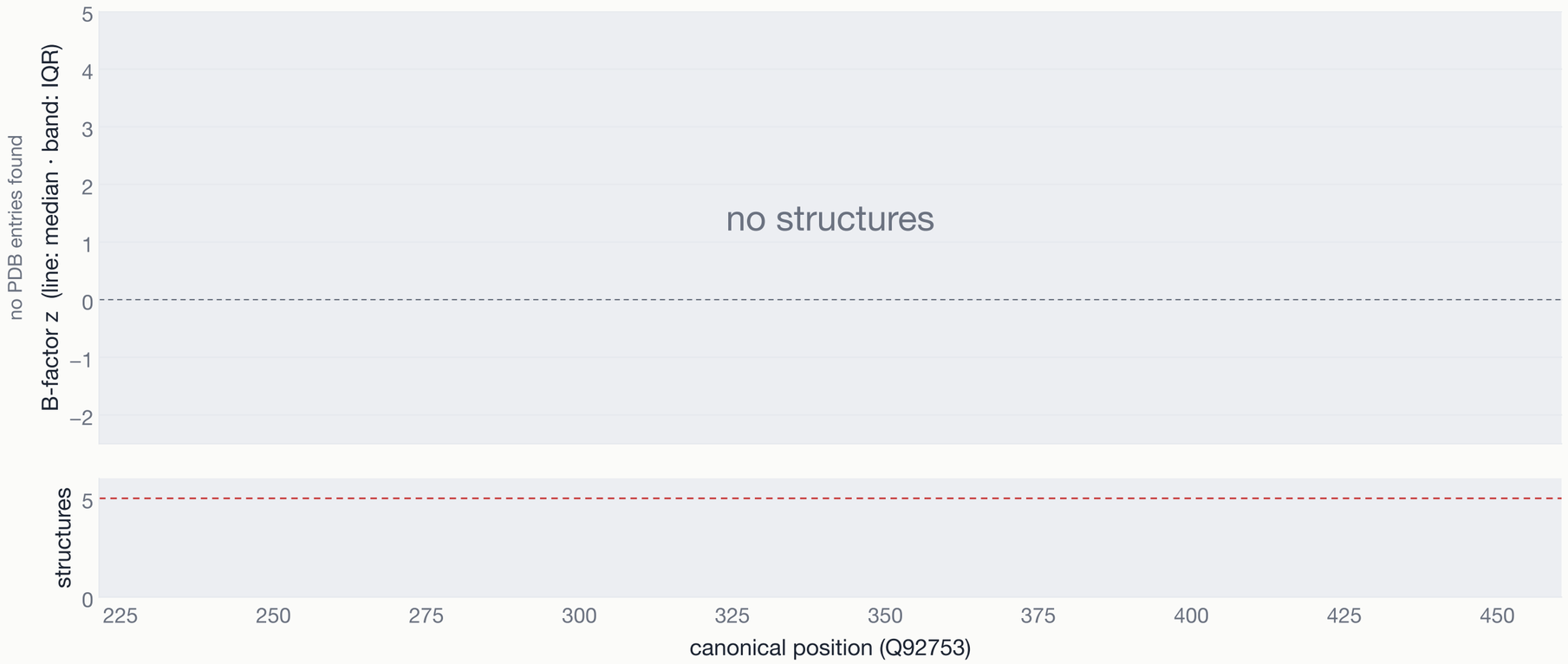
LIGANDS

–

PARTNERS / COFACTORS

–

no professor helix map



BETWEEN-PDB HETEROGENEITY

not assessable

No structures

CAVEATS

- No professor helix map: no canonical elements.

RORγ NR1F3 · 162 usable PDBs

mostly holo · ligand-diverse · partly partner-bound · high IQR heterogeneity

STRUCTURES

HOLO 155

APO 4 AMBIG. 3

162 found · 162 with coordinates · 100% of LBD observed (median)

LIGANDS

147 unique

4F1 6 · L3E 5 · MJE 5

154 ligand–partner contexts

dominant: APO + standalone (2%)

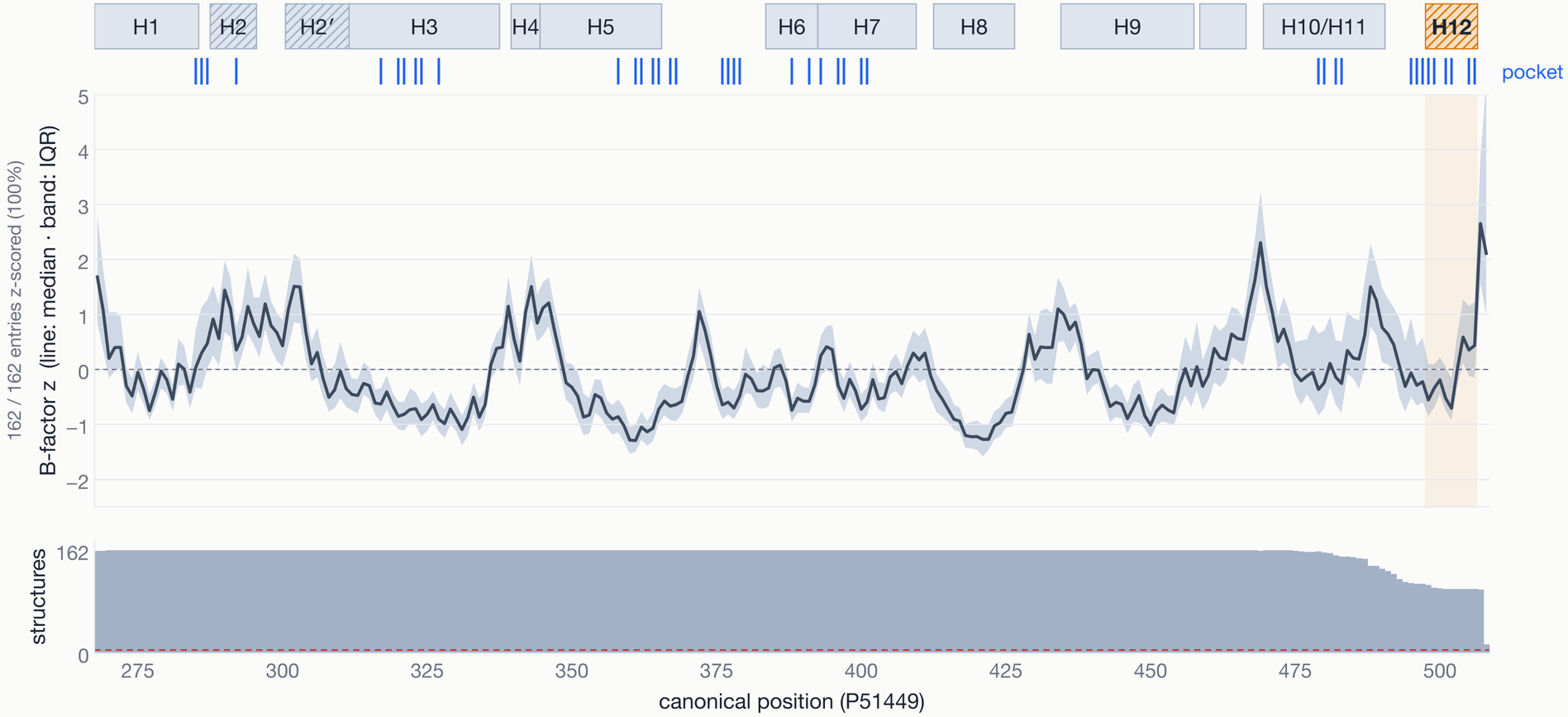
PARTNERS / COFACTORS

other 26 | standalone 110

coregulator 26

most common: nuclear receptor-interacting protein 1

10 unique partner identities



BETWEEN-PDB HETEROGENEITY

HIGH

median IQR	0.72
90th percentile	1.27
supported positions	241 / 241

CONSTRUCTS & CRYSTALS

32 construct patterns · 49 crystal forms

2.21 Å median (1.91–2.50)

substitution status uncertain: 12%

14 fusion/chimera

LXRβ NR1H2 · 25 usable PDBs

holo only · ligand-diverse · partly partner-bound · high IQR heterogeneity

STRUCTURES

HOLO 25 — no APO

25 found · 25 with coordinates · 98% of LBD observed (median)

LIGANDS

23 unique

444 3 · 965 2 · CO1 1

23 ligand–partner contexts

dominant: 444 + standalone (12%)

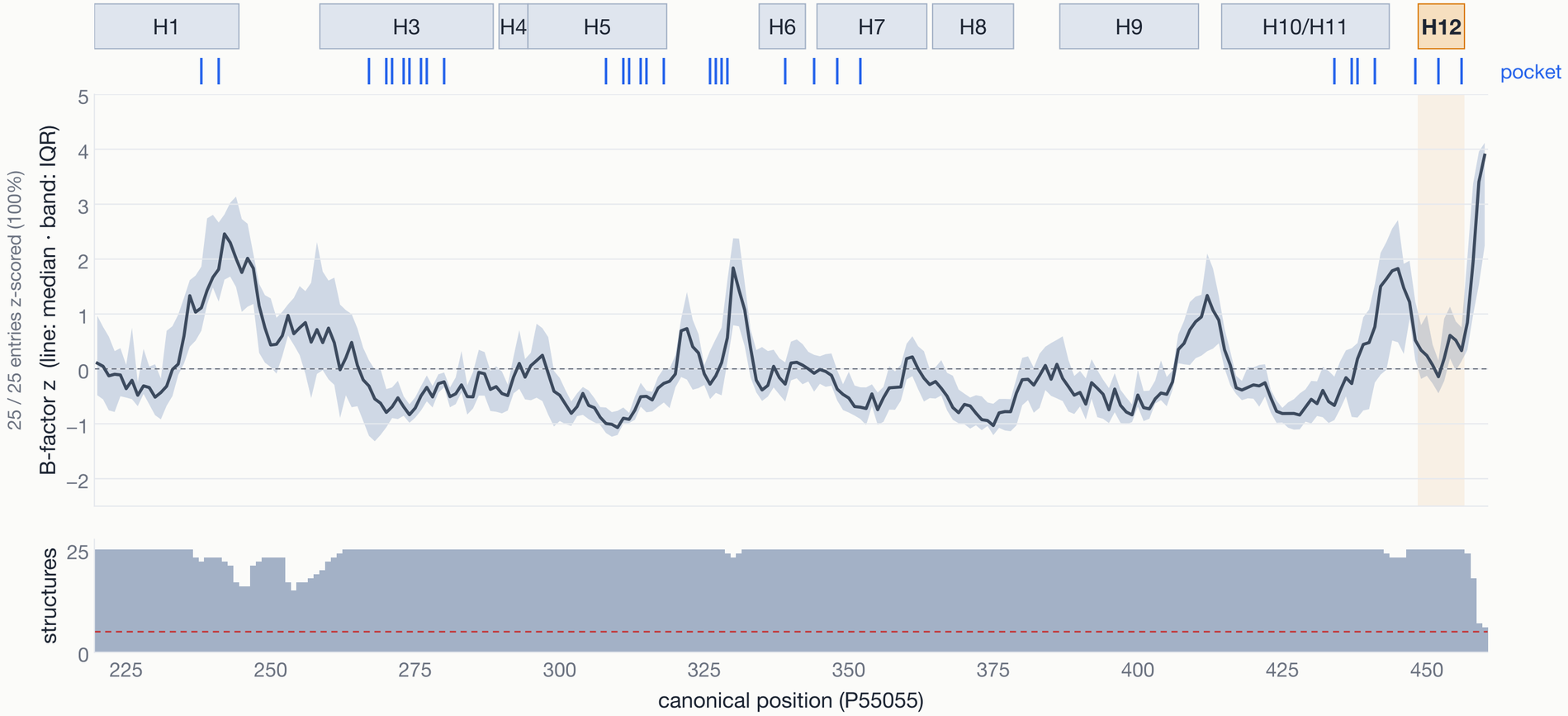
PARTNERS / COFACTORS

NR partner 5 | standalone 16

coregulator 4

most common: retinoic acid receptor rxr-beta

4 unique partner identities



BETWEEN-PDB HETEROGENEITY

HIGH

median IQR	0.76
90th percentile	1.58
supported positions	241 / 241

CONSTRUCTS & CRYSTALS

8 construct patterns · 19 crystal forms

2.50 Å median (2.30–2.80)

engineered mutants: 36% (top R263S)

4 internal deletion

CAVEATS

- Very heterogeneous crystal forms (19 forms in 25 structures).

LXRα NR1H3 · 7 usable PDBs

holo only · ligand-diverse · partner-rich · low IQR heterogeneity

STRUCTURES

HOLO 7 — no APO

7 found · 7 with coordinates · 93% of LBD observed (median)

LIGANDS

7 unique

444 1 · 965 1 · O90 1

7 ligand–partner contexts

dominant: 444 + NCOA2 + RXRβ (14%)

PARTNERS / COFACTORS

coregulator 6

NR partner 1

most common: nuclear receptor coactivator 1

3 unique partner identities



BETWEEN-PDB HETEROGENEITY

LOW

median IQR	0.23
90th percentile	0.46
supported positions	220 / 241

CONSTRUCTS & CRYSTALS

2 construct patterns · 3 crystal forms

2.40 Å median (2.13–2.75)

FXR NR1H4 · 91 usable PDBs

mostly holo · ligand-diverse · partner-rich · high IQR heterogeneity

STRUCTURES

HOLO 89

APO 2

92 found · 91 with coordinates · 100% of LBD observed (median)

LIGANDS

84 unique

064 2 · 33Y 2 · 643 2

90 ligand–partner contexts

dominant: 643 + NCOA1 (2%)

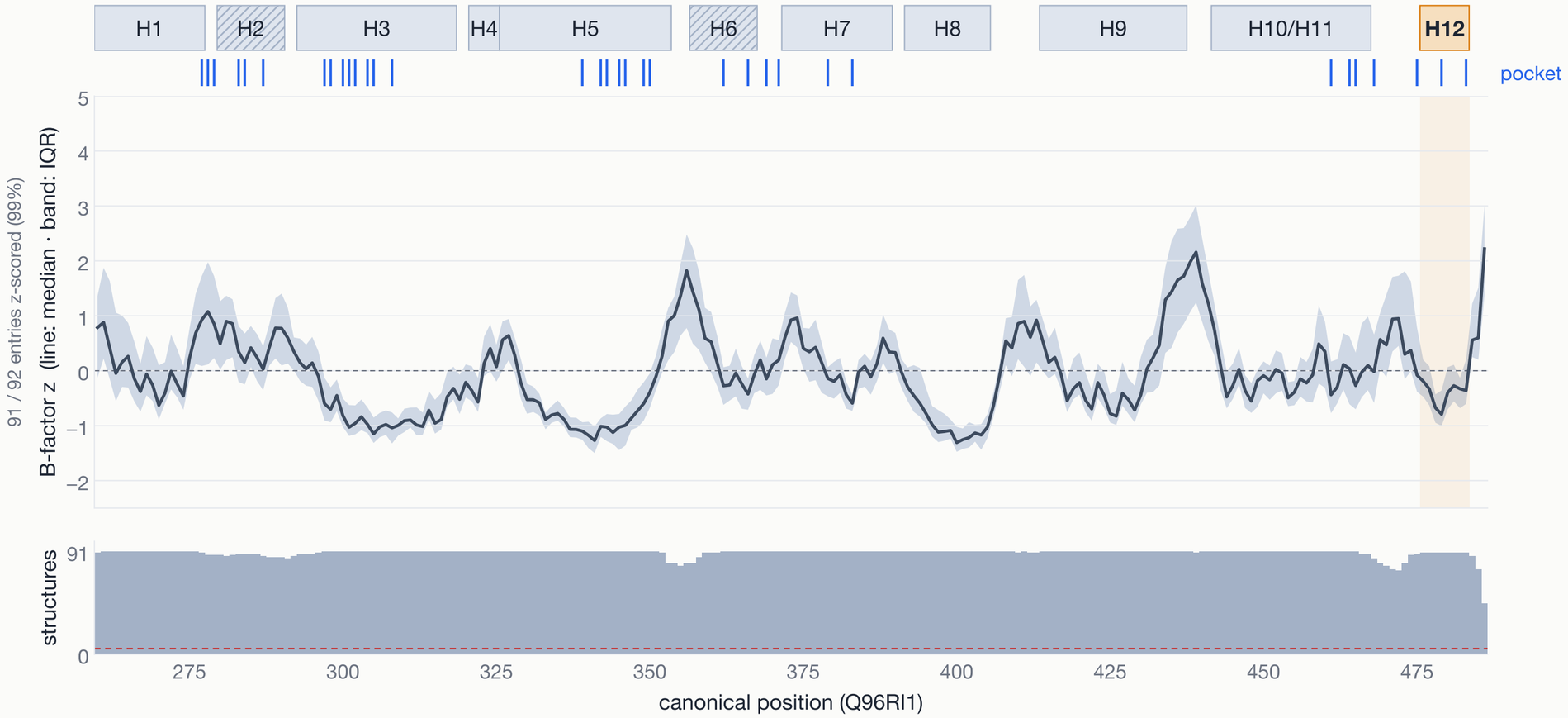
PARTNERS / COFACTORS

coregulator 79

NR partner 4 other 2 standalone 6

most common: coactivator peptide src-1 hd3

6 unique partner identities



BETWEEN-PDB HETEROGENEITY

HIGH

median IQR	0.74
90th percentile	1.43
supported positions	227 / 227

CONSTRUCTS & CRYSTALS

13 construct patterns · 40 crystal forms

2.26 Å median (1.96–2.60)

engineered mutants: 23% (top E291A)

substitution status uncertain: 50%

VDR NR1I1 · 48 usable PDBs

holo only · ligand-diverse · mostly standalone · low IQR heterogeneity

STRUCTURES

HOLO 48 — no APO

52 found · 52 with coordinates · 83% of LBD observed (median)

LIGANDS

42 unique

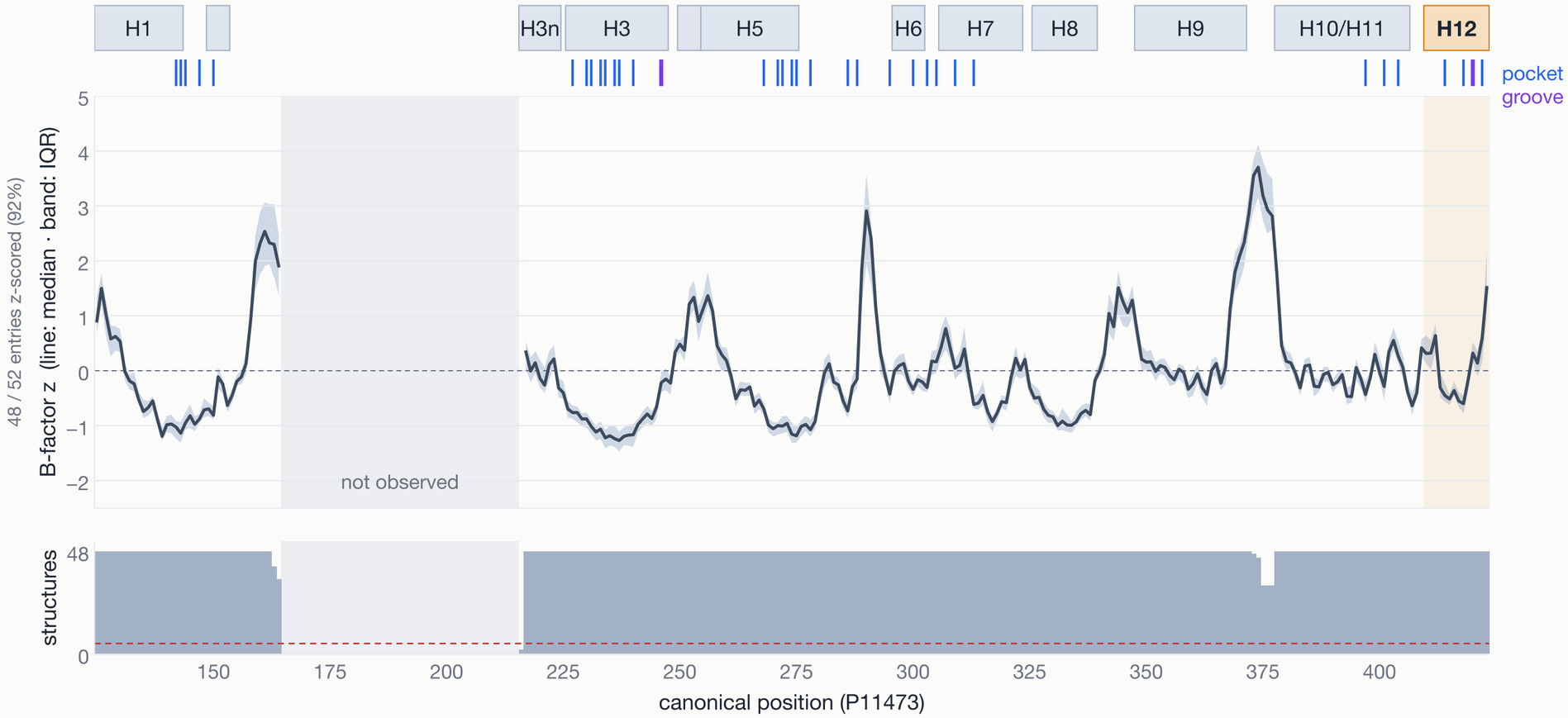
VDX 4 · TEJ 2 · CA9 2

42 ligand–partner contexts

dominant: calcitriol + standalone (8%)

PARTNERS / COFACTORS

Standalone: 100%



BETWEEN-PDB HETEROGENEITY

LOW

median IQR	0.34
90th percentile	0.63
supported positions	247 / 299

CONSTRUCTS & CRYSTALS

4 construct patterns · 3 crystal forms

1.90 Å median (1.75–2.10)

46 internal deletion

CAVEATS

- 46 of 48 constructs carry an internal deletion.

PXR NR1I2 · 80 usable PDBs

mixed apo/holo · ligand-diverse · partly partner-bound · low IQR heterogeneity

STRUCTURES

HOLO 69

APO 11

80 found · 80 with coordinates · 93% of LBD observed (median)

LIGANDS

56 unique

SRL 6 · EST 4 · WU6 4

63 ligand–partner contexts

dominant: APO + standalone (11%)

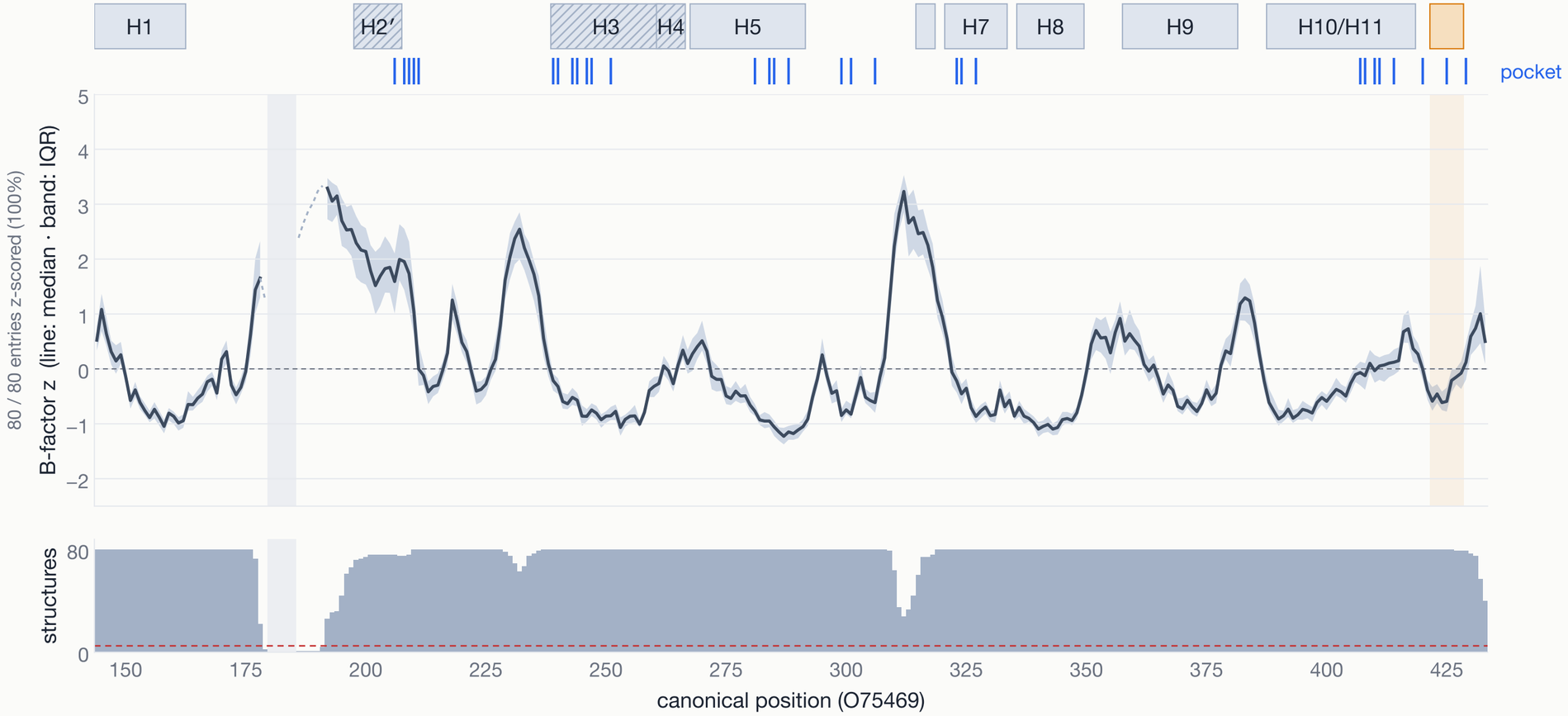
PARTNERS / COFACTORS

standalone 71

coregulator 5 NR partner 2 other 2

most common: nuclear receptor coactivator 1

3 unique partner identities



BETWEEN-PDB HETEROGENEITY

LOW

median IQR	0.36
90th percentile	0.79
supported positions	277 / 290

CONSTRUCTS & CRYSTALS

8 construct patterns · 10 crystal forms

2.33 Å median (2.15–2.68)

25 fusion/chimera

CAVEATS

- Fusion/chimera constructs in 31% of structures.

CAR NR1I3 · 2 usable PDBs

mixed apo/holo · one dominant ligand · partner-rich · IQR not assessable

STRUCTURES

HOLO 1 | APO 1

2 found · 2 with coordinates · 98% of LBD observed (median)

LIGANDS

1 unique · CI2 in 100%

CI2 1

2 ligand–partner contexts

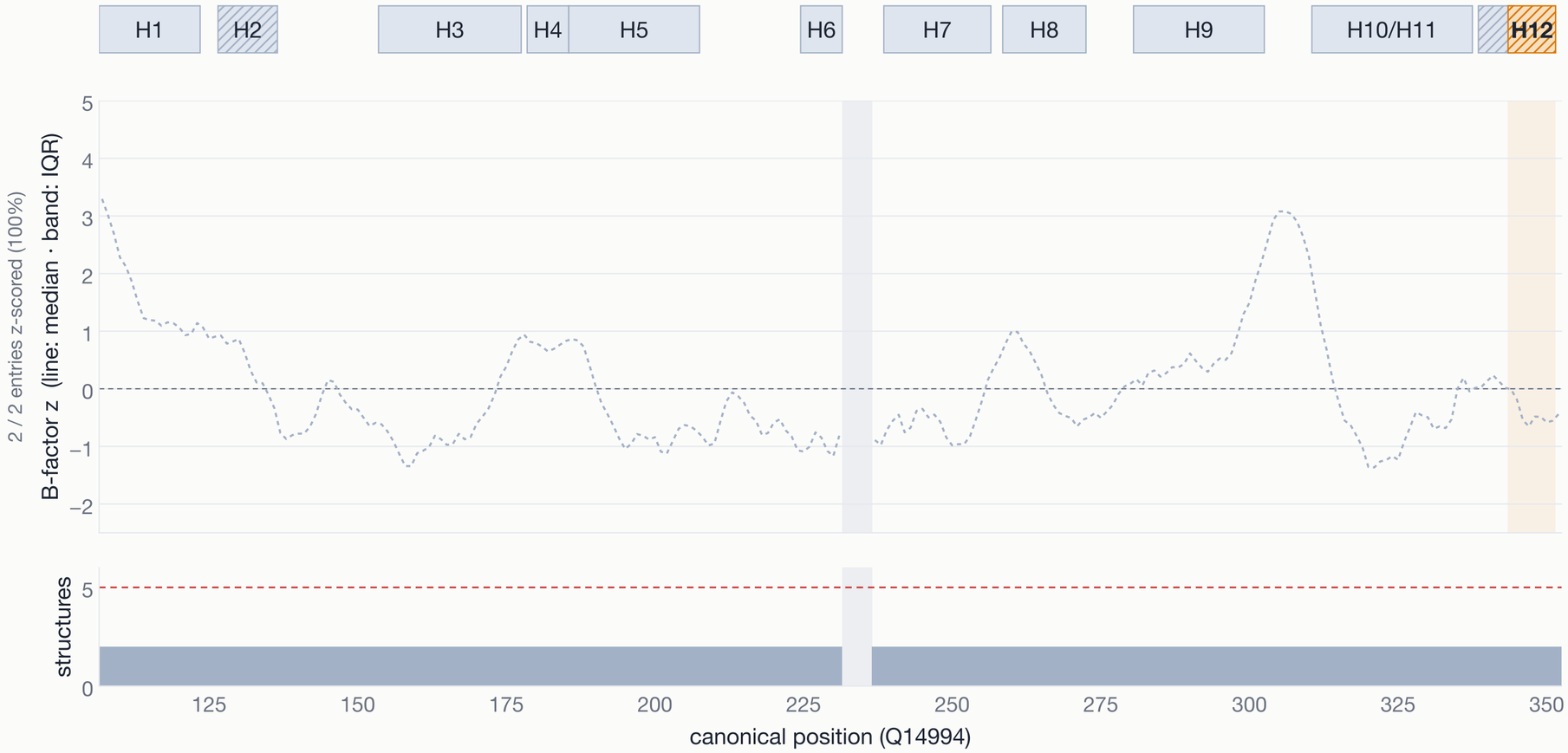
dominant: CI2 + NCOA1 + RXRα (50%)

PARTNERS / COFACTORS

NR partner 2

most common: retinoic acid receptor rxr-alpha

2 unique partner identities



BETWEEN-PDB HETEROGENEITY

not assessable

Only 2 usable structures

CONSTRUCTS & CRYSTALS

1 construct pattern · 1 crystal form

2.65 Å median

2 internal deletion

CAVEATS

- 2 of 2 constructs carry an internal deletion.

HNF4α NR2A1 · 7 usable PDBs

mixed apo/holo · one dominant ligand · partner-rich · moderate IQR heterogeneity

STRUCTURES

HOLO 6 | APO 1

8 found · 7 with coordinates · 97% of LBD observed (median)

LIGANDS

4 unique · MYR in 50%

MYR 3 · DAO 1 · PLM 1

7 ligand–partner contexts

dominant: myristic acid + NCOA1 (14%)

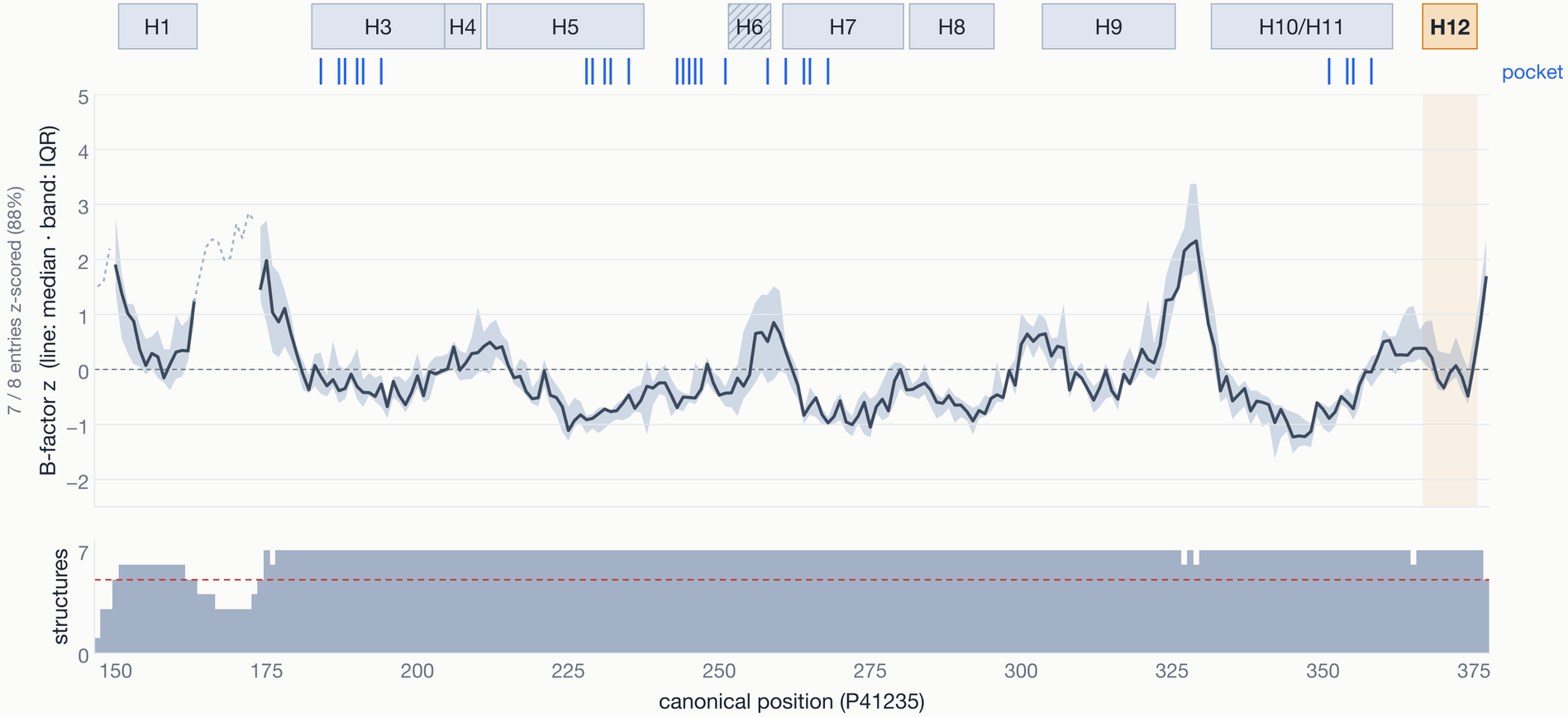
PARTNERS / COFACTORS

coregulator 5 | other 1

standalone 1

most common: nuclear receptor coactivator 2

4 unique partner identities



BETWEEN-PDB HETEROGENEITY

MODERATE

median IQR	0.54
90th percentile	1.06
supported positions	218 / 231

CONSTRUCTS & CRYSTALS

5 construct patterns · 7 crystal forms

2.78 Å median (2.15–3.04)

HNF4γ NR2A2 · 1 usable PDB

holo only · one dominant ligand · mostly standalone · IQR not assessable

STRUCTURES

HOLO 1 no APO

1 found · 1 with coordinates · 96% of LBD observed (median)

LIGANDS

1 unique · PLM in 100%

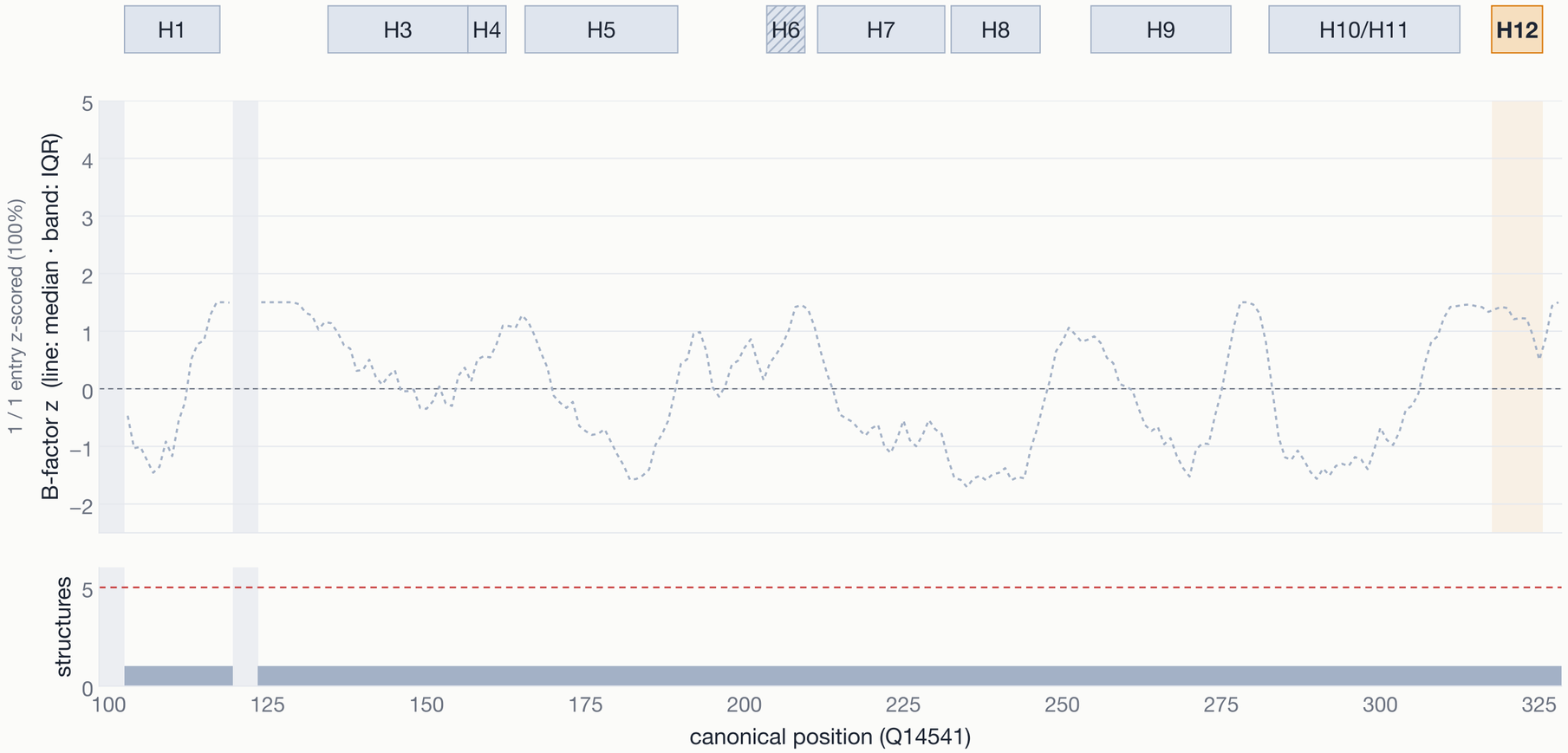
PLM 1

1 ligand–partner context

dominant: palmitic acid + standalone (100%)

PARTNERS / COFACTORS

Standalone: 100%



BETWEEN-PDB HETEROGENEITY

not assessable

Only 1 usable structure

CONSTRUCTS & CRYSTALS

1 construct pattern

2.70 Å median

RXRα NR2B1 · 94 usable PDBs

mixed apo/holo · ligand-diverse · partner-rich · high IQR heterogeneity

STRUCTURES

HOLO 82

APO 12

110 found · 95 with coordinates · 92% of LBD observed (median)

LIGANDS

62 unique

9CR 17 · REA 3 · BM6 2

77 ligand–partner contexts

dominant: APO + standalone (5%)

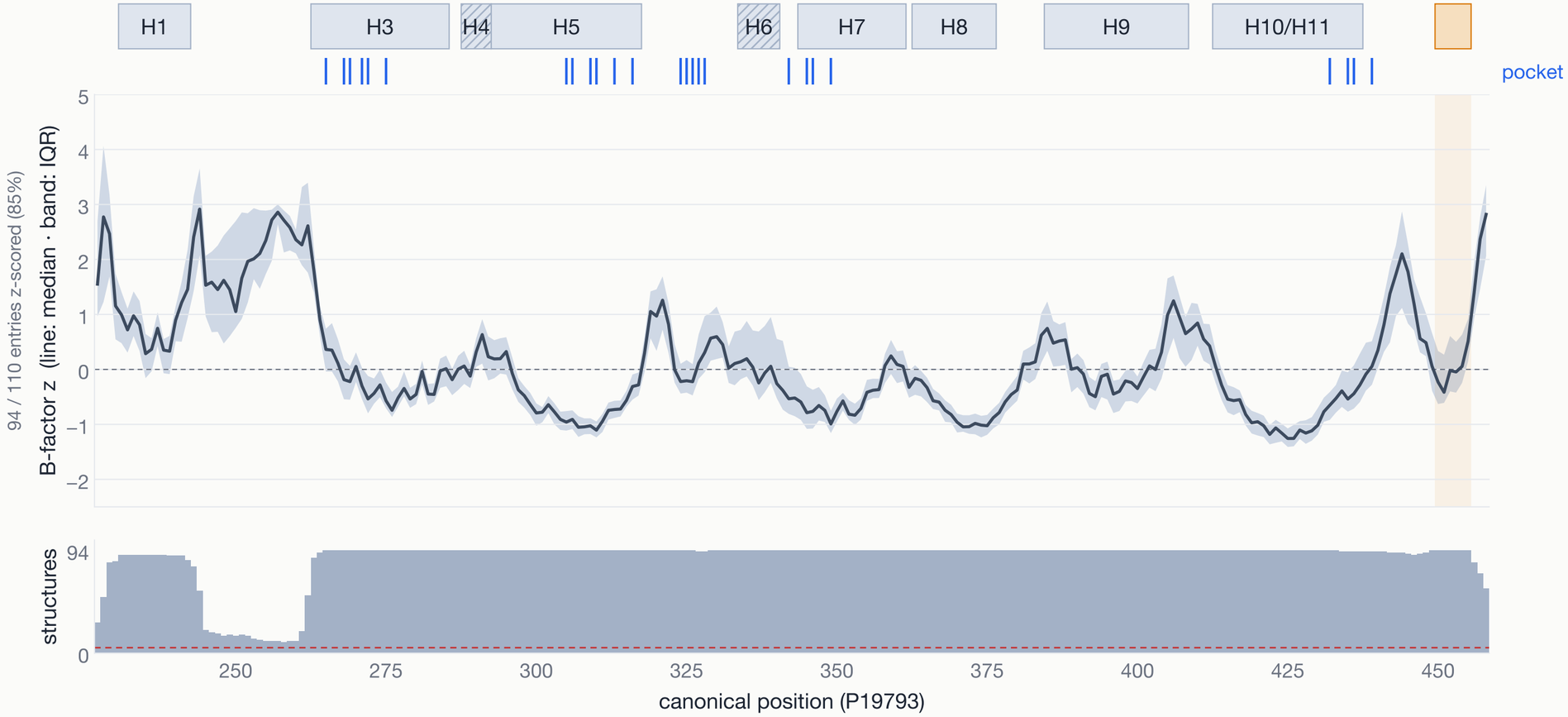
PARTNERS / COFACTORS

coregulator 50 | NR partner 24

other 8 standalone 12

most common: nuclear receptor coactivator 2

15 unique partner identities



BETWEEN-PDB HETEROGENEITY

HIGH

median IQR	0.61
90th percentile	1.25
supported positions	232 / 232

CONSTRUCTS & CRYSTALS

16 construct patterns · 41 crystal forms

2.30 Å median (2.00–2.70)

RXRβ NR2B2 · 7 usable PDBs

mixed apo/holo · ligand-diverse · partner-rich · moderate IQR heterogeneity

STRUCTURES

HOLO 3 | APO 4

7 found · 7 with coordinates · 89% of LBD observed (median)

LIGANDS

3 unique

LG2 1 · MEI 1 · PLM 1

5 ligand–partner contexts

dominant: APO + LXRβ (43%)

PARTNERS / COFACTORS

NR partner 5

coregulator 1 standalone 1

most common: oxysterols receptor lxr-beta

3 unique partner identities



BETWEEN-PDB HETEROGENEITY

MODERATE

median IQR	0.55
90th percentile	1.03
supported positions	207 / 234

CONSTRUCTS & CRYSTALS

4 construct patterns · 5 crystal forms

2.61 Å median (2.60–2.75)

substitution status uncertain: 29%

1 fusion/chimera

RXRγ NR2B3 · 2 usable PDBs

mixed apo/holo · one dominant ligand · partner-rich · IQR not assessable

STRUCTURES

HOLO 1 | APO 1

2 found · 2 with coordinates · 90% of LBD observed (median)

LIGANDS

1 unique · PLM in 100%

PLM 1

2 ligand–partner contexts

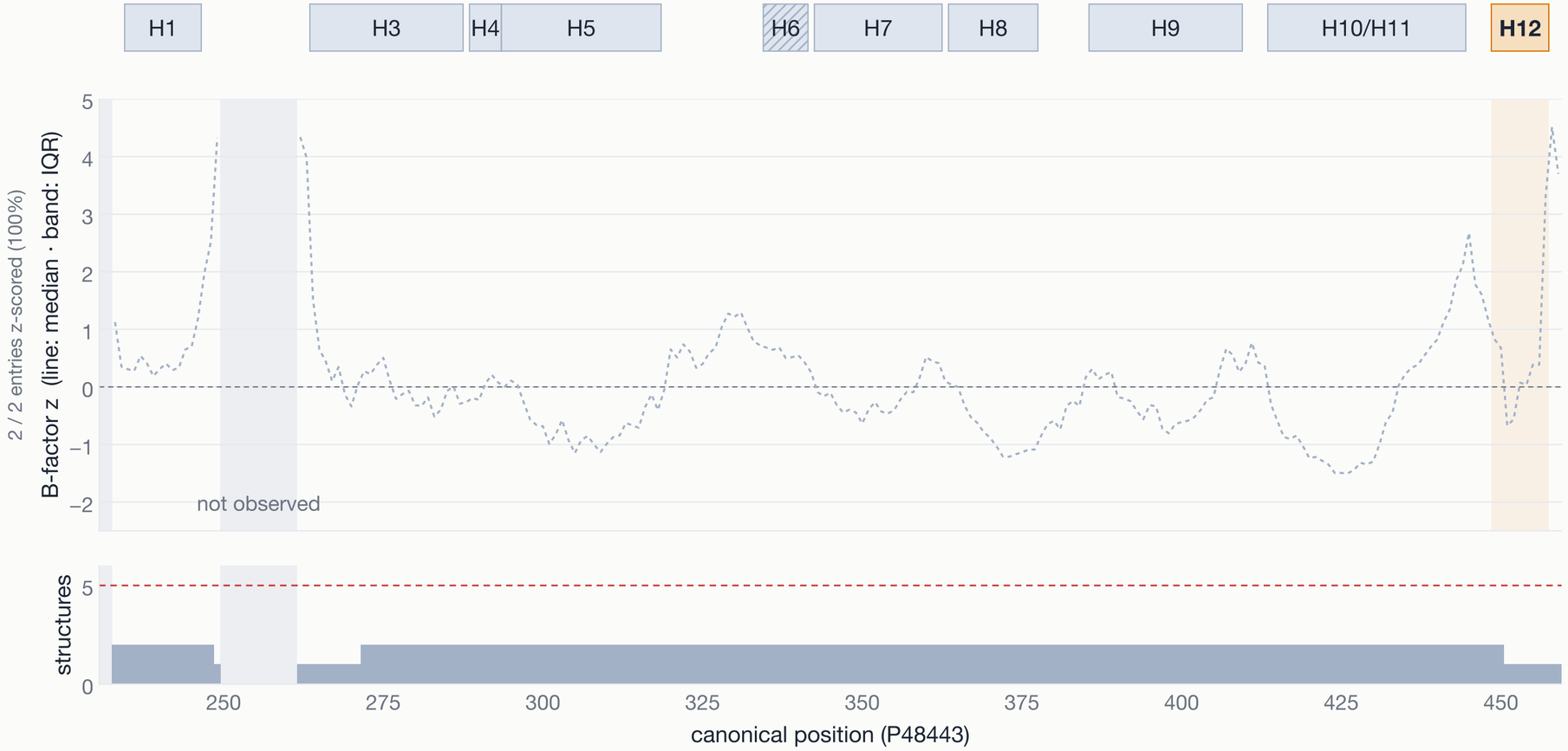
dominant: APO + standalone (50%)

PARTNERS / COFACTORS

coregulator 1 | standalone 1

most common: nuclear receptor coactivator 2

1 unique partner identity



BETWEEN-PDB HETEROGENEITY

not assessable

Only 2 usable structures

CONSTRUCTS & CRYSTALS

2 construct patterns · 2 crystal forms

2.22 Å median

TR2 NR2C1 · 0 usable PDBs

no usable LBD structures

STRUCTURES

No usable LBD structures

0 entries found

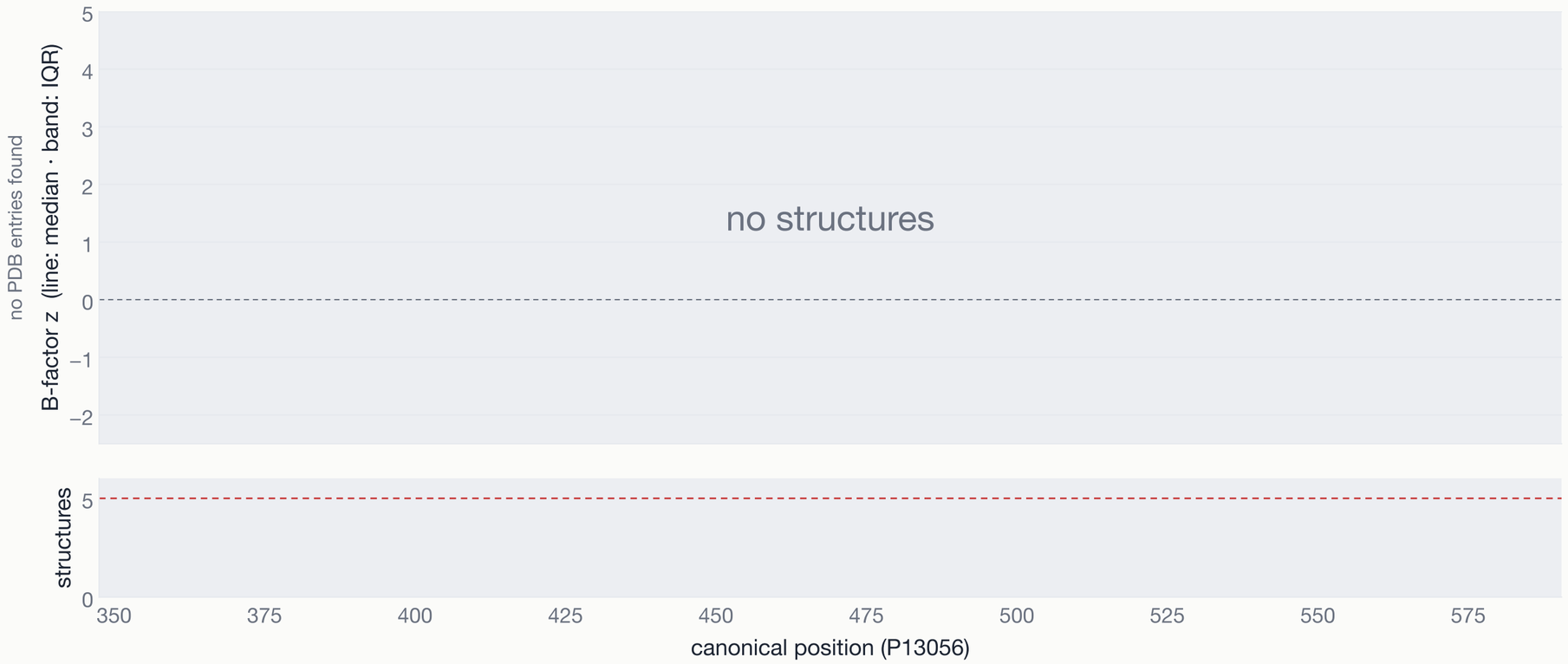
LIGANDS

–

PARTNERS / COFACTORS

–

no professor helix map



BETWEEN-PDB HETEROGENEITY

not assessable

No structures

CAVEATS

- No professor helix map: no canonical elements.

TR4 NR2C2 · 2 usable PDBs

apo only · no bound ligands · partner-rich · IQR not assessable

STRUCTURES

APO 2

5 found · 2 with coordinates · 90% of LBD observed (median)

LIGANDS

no bound ligands

2 ligand–partner contexts

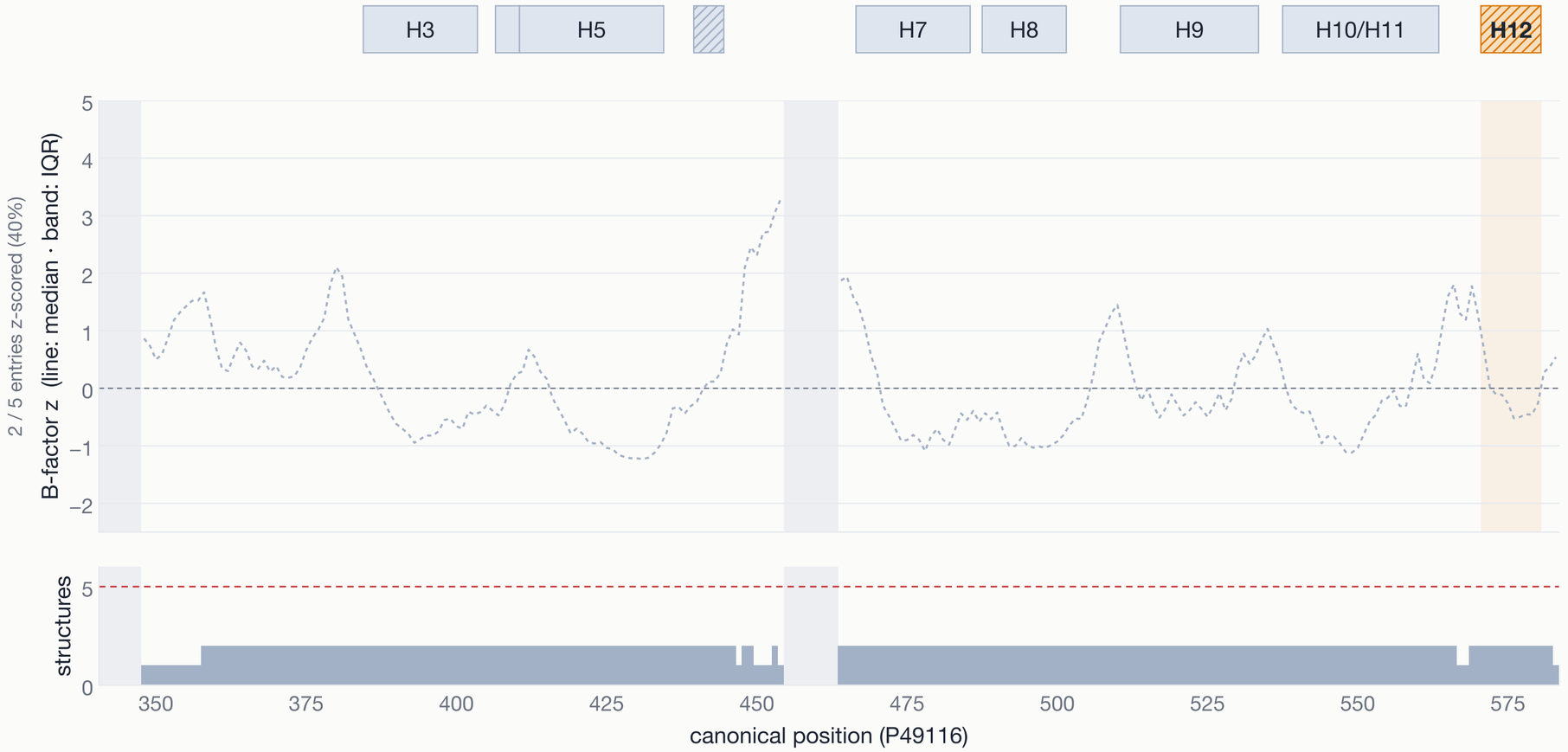
dominant: APO + standalone (50%)

PARTNERS / COFACTORS

other 1 | standalone 1

most common: juxtaposed with another zinc finger…

1 unique partner identity



BETWEEN-PDB HETEROGENEITY

not assessable

Only 2 usable structures

CONSTRUCTS & CRYSTALS

2 construct patterns · 2 crystal forms

2.43 Å median

engineered mutants: 50% (top K520A)

TLX NR2E1 · 1 usable PDB

apo only · no bound ligands · partner-rich · IQR not assessable

STRUCTURES

APO 1

1 found · 1 with coordinates · 86% of LBD observed (median)

LIGANDS

no bound ligands

1 ligand–partner context

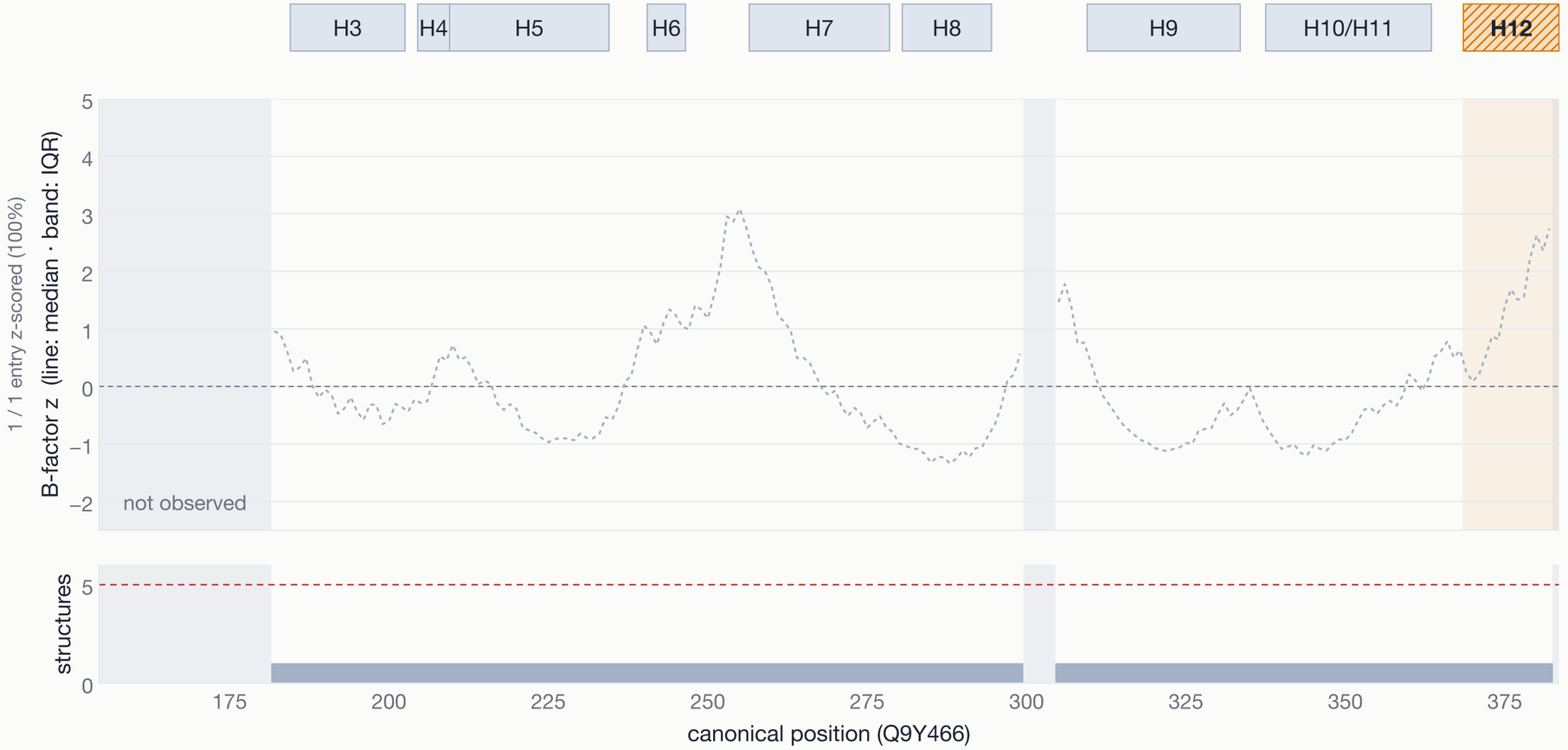
dominant: APO + atrophin/grunge (100%)

PARTNERS / COFACTORS

other 1

most common: atrophin/grunge

1 unique partner identity



BETWEEN-PDB HETEROGENEITY

not assessable

Only 1 usable structure

CONSTRUCTS & CRYSTALS

1 construct pattern

3.55 Å median

engineered mutants: 100% (top K257R)

1 fusion/chimera · 1 internal deletion

CAVEATS

- 1 of 1 constructs carry an internal deletion.
- Fusion/chimera constructs in 100% of structures.

PNR NR2E3 · 1 usable PDB

apo only · no bound ligands · mostly standalone · IQR not assessable

STRUCTURES

APO 1

1 found · 1 with coordinates · 78% of LBD observed (median)

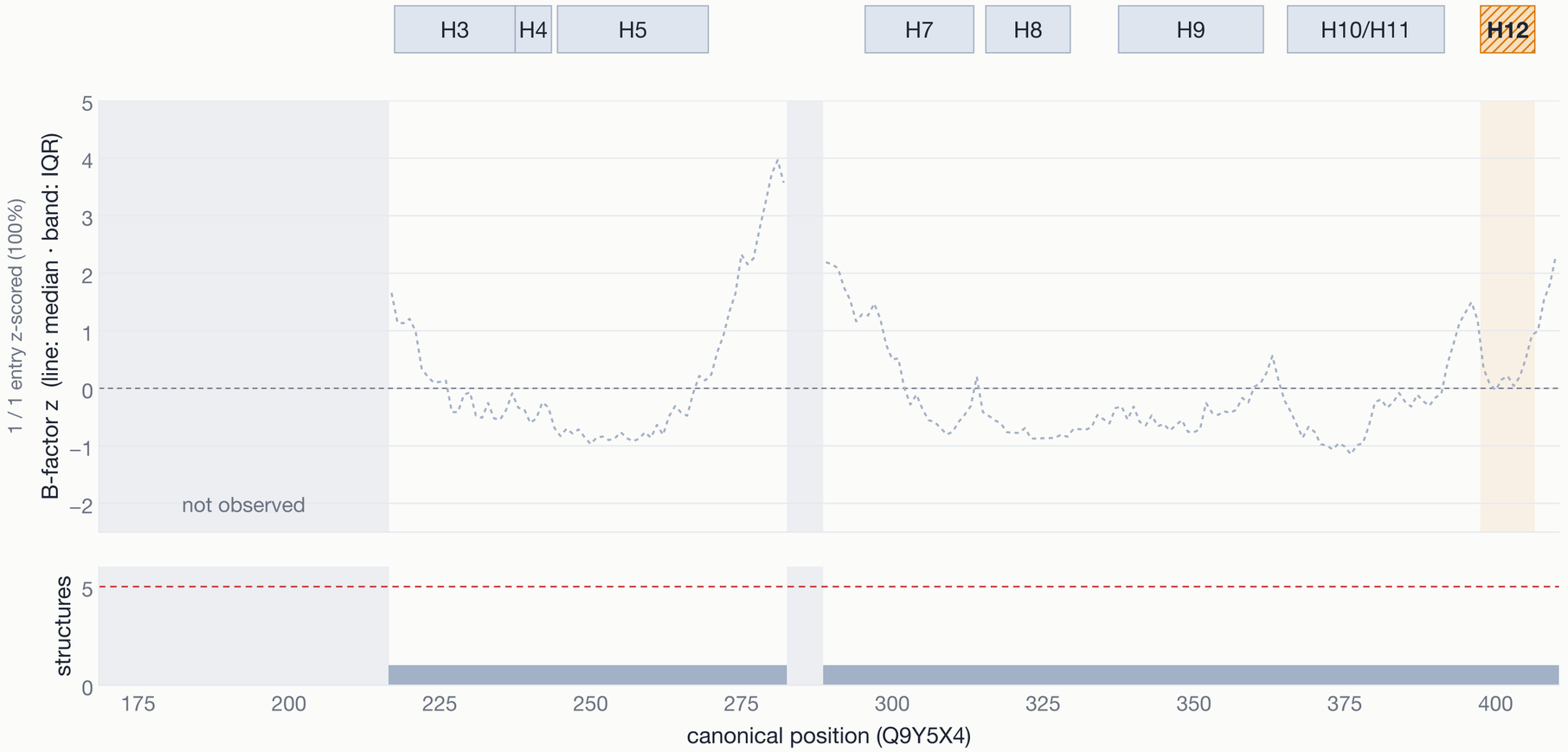
LIGANDS

no bound ligands

PARTNERS / COFACTORS

Standalone: 100%

1 ligand–partner context
dominant: APO + standalone (100%)



BETWEEN-PDB HETEROGENEITY

not assessable

Only 1 usable structure

CONSTRUCTS & CRYSTALS

1 construct pattern
2.70 Å median
engineered mutants: 100% (top C405S)
1 fusion/chimera · 1 internal deletion

CAVEATS

- 1 of 1 constructs carry an internal deletion.
- Fusion/chimera constructs in 100% of structures.

COUP-TFI NR2F1 · 0 usable PDBs

no usable LBD structures

STRUCTURES

No usable LBD structures

1 entry found (not X-ray (NMR/EM) (1))

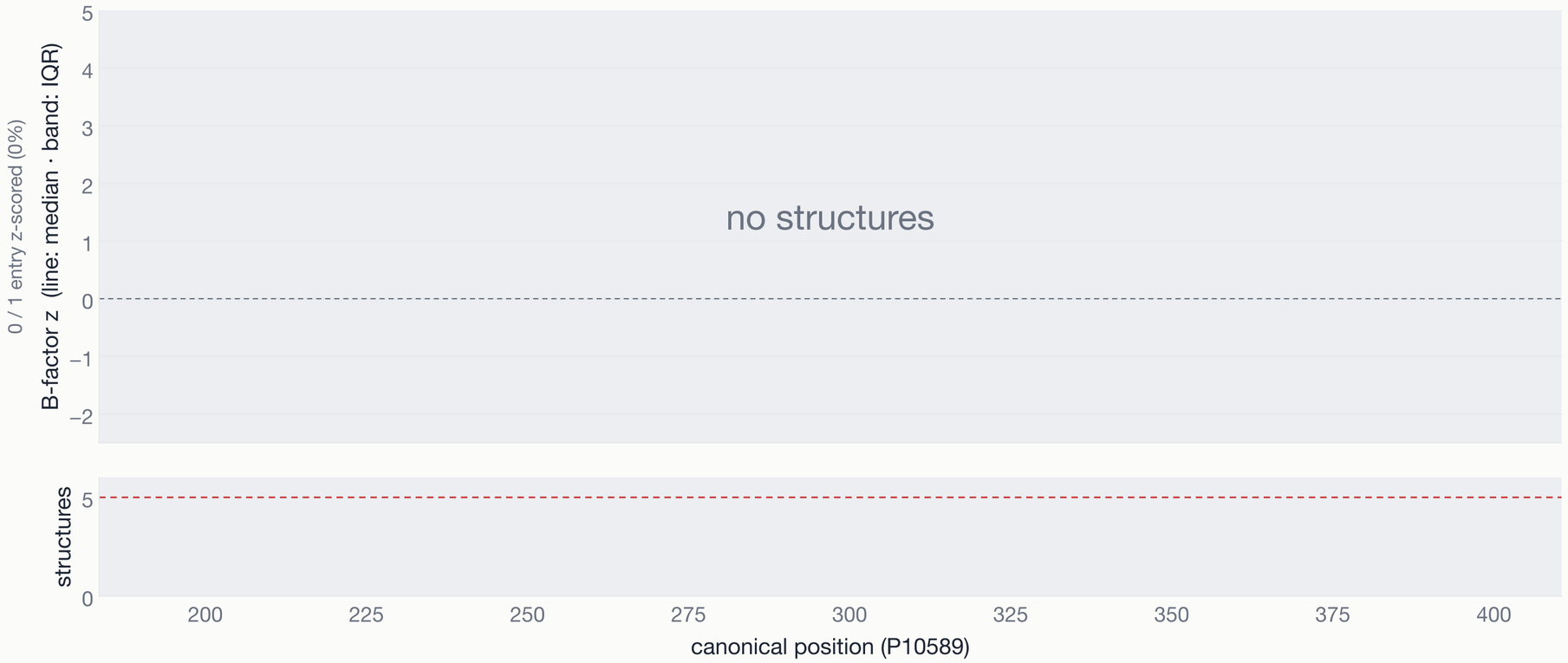
LIGANDS

–

PARTNERS / COFACTORS

–

no professor helix map



BETWEEN-PDB HETEROGENEITY

not assessable

No structures

CAVEATS

- No professor helix map: no canonical elements.

COUP-TFII NR2F2 · 1 usable PDB

apo only · no bound ligands · mostly standalone · IQR not assessable

STRUCTURES

APO 1

1 found · 1 with coordinates · 87% of LBD observed (median)

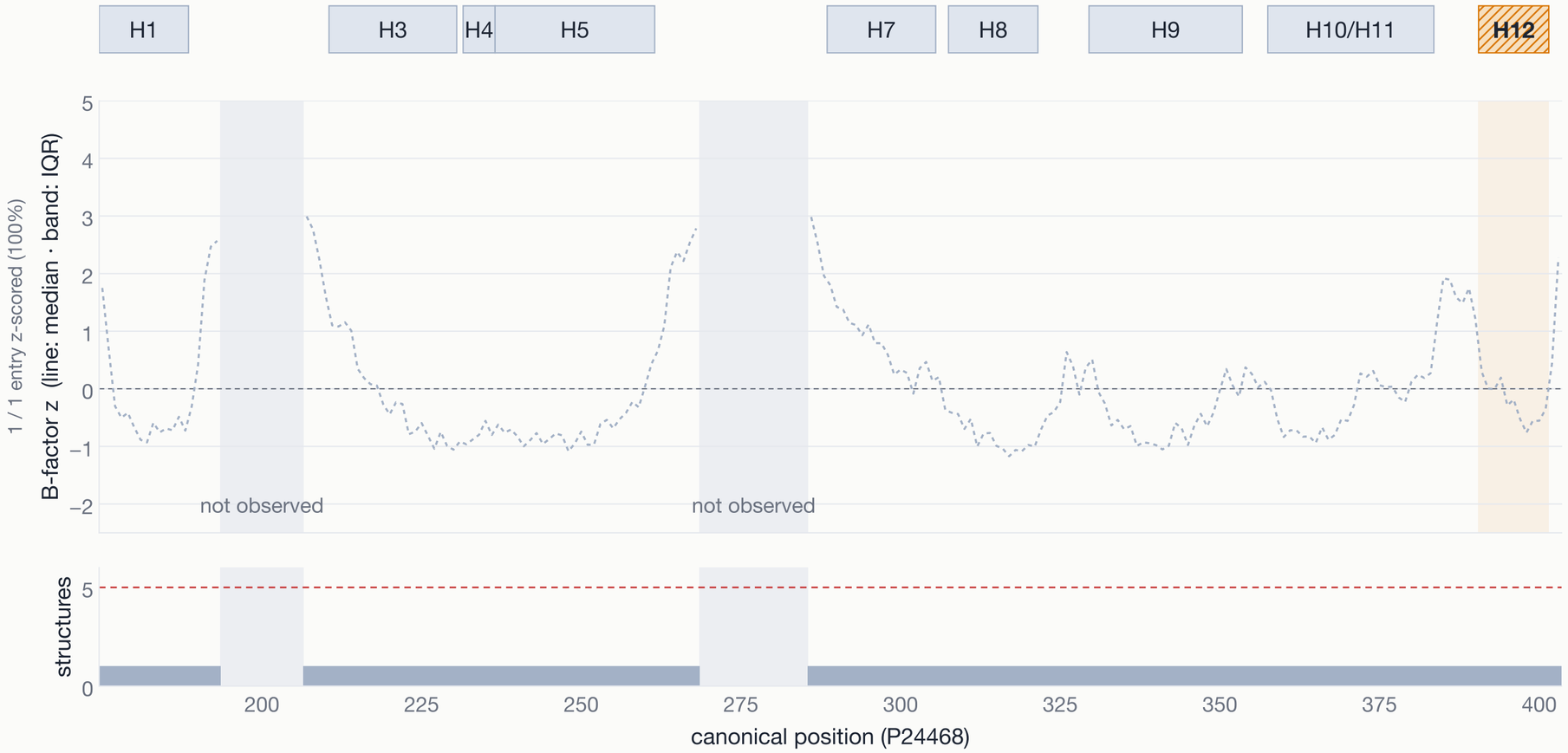
LIGANDS

no bound ligands

1 ligand–partner context
dominant: APO + standalone (100%)

PARTNERS / COFACTORS

Standalone: 100%



BETWEEN-PDB HETEROGENEITY

not assessable

Only 1 usable structure

CONSTRUCTS & CRYSTALS

1 construct pattern

1.48 Å median

EAR-2 NR2F6 · 1 usable PDB

apo only · no bound ligands · partner-rich · IQR not assessable

STRUCTURES

APO 1

1 found · 1 with coordinates · 80% of LBD observed (median)

LIGANDS

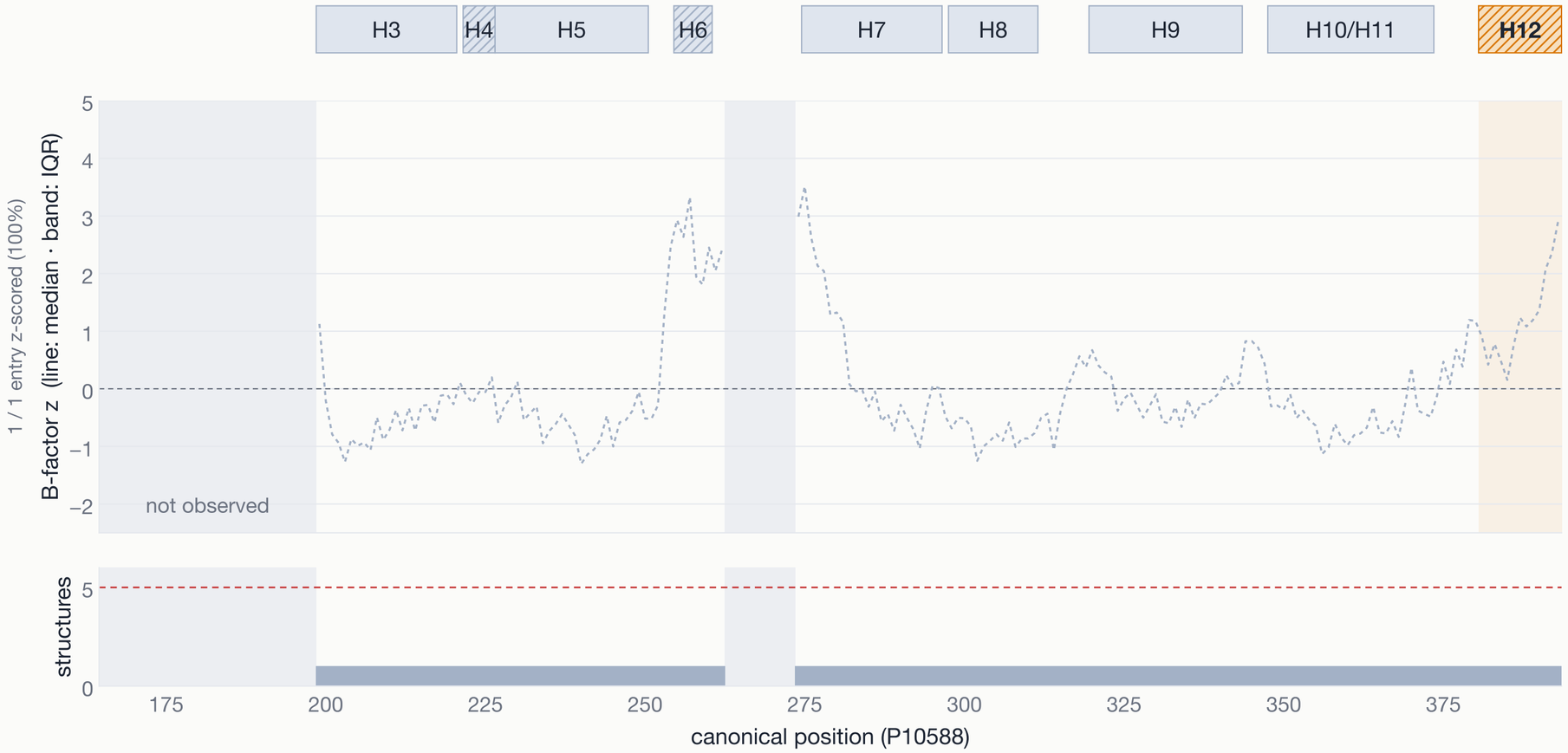
no bound ligands

1 ligand–partner context
dominant: APO + NSD1 (100%)

PARTNERS / COFACTORS

other 1

most common: histone-lysine n-methyltransferase, h3…
1 unique partner identity



BETWEEN-PDB HETEROGENEITY

not assessable

Only 1 usable structure

CONSTRUCTS & CRYSTALS

1 construct pattern
2.60 Å median
1 fusion/chimera · 1 internal deletion

CAVEATS

- 1 of 1 constructs carry an internal deletion.
- Fusion/chimera constructs in 100% of structures.

ERα NR3A1 · 351 usable PDBs

mostly holo · ligand-diverse · partner-rich · moderate IQR heterogeneity

STRUCTURES

HOLO 343

APO 8

492 found · 351 with coordinates · 97% of LBD observed (median)

LIGANDS

293 unique

EST 23 · OHT 7 · RAL 6

307 ligand–partner contexts

dominant: estradiol + NCOA2 (3%)

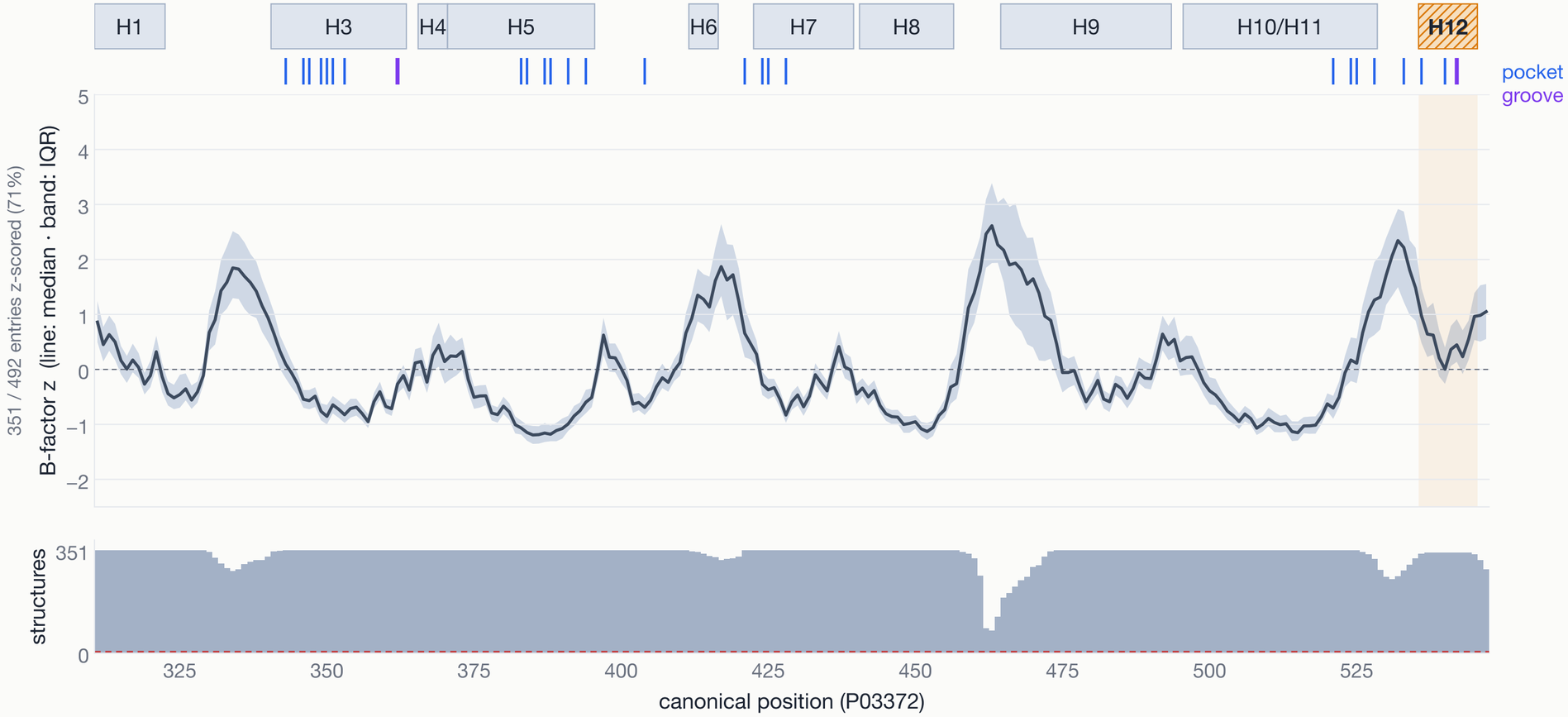
PARTNERS / COFACTORS

coregulator 181 | standalone 137

other 33

most common: nuclear receptor coactivator 2

9 unique partner identities



BETWEEN-PDB HETEROGENEITY

MODERATE

median IQR	0.46
90th percentile	1.21
supported positions	237 / 237

CONSTRUCTS & CRYSTALS

35 construct patterns · 69 crystal forms

2.10 Å median (1.86–2.30)

engineered mutants: 88% (top Y537S)

ERβ NR3A2 · 39 usable PDBs

holo only · ligand-diverse · partner-rich · moderate IQR heterogeneity

STRUCTURES

HOLO 39 — no APO

39 found · 39 with coordinates · 96% of LBD observed (median)

LIGANDS

33 unique

EST 5 · GEN 2 · B81 2

36 ligand–partner contexts

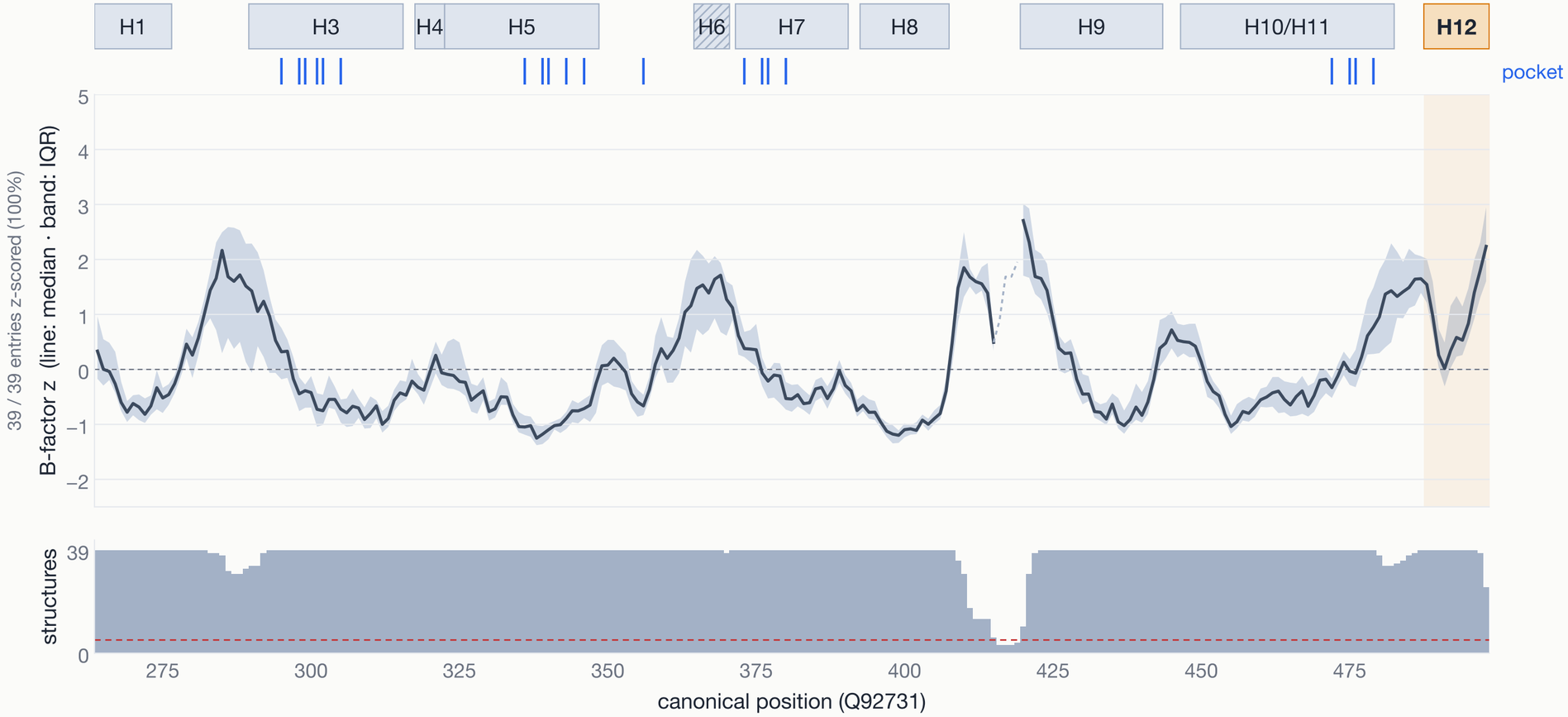
dominant: estradiol + NCOA1 (5%)

PARTNERS / COFACTORS

coregulator 20 | other 7 | standalone 12

most common: steroid receptor coactivator-1

4 unique partner identities



BETWEEN-PDB HETEROGENEITY

MODERATE

median IQR	0.54
90th percentile	1.18
supported positions	231 / 235

CONSTRUCTS & CRYSTALS

12 construct patterns · 19 crystal forms

2.20 Å median (2.06–2.37)

ERRα NR3B1 · 5 usable PDBs

mixed apo/holo · ligand-diverse · partner-rich · high IQR heterogeneity

STRUCTURES

HOLO 3 | APO 2

6 found · 5 with coordinates · 93% of LBD observed (median)

LIGANDS

3 unique

047 1 · 5FB 1 · HVO 1

4 ligand–partner contexts

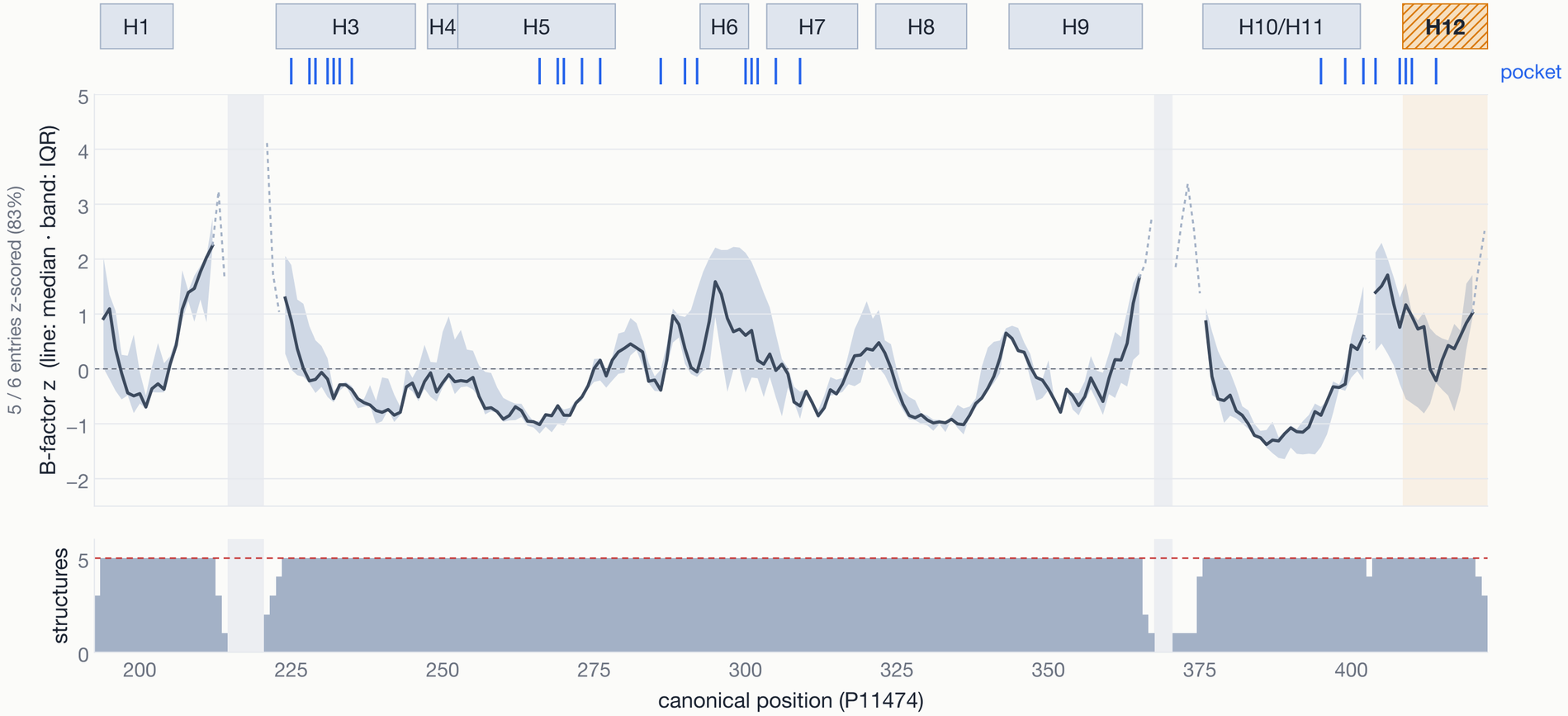
dominant: APO + PPARGC1A (40%)

PARTNERS / COFACTORS

coregulator 3 | standalone 2

most common: peroxisome proliferator-activated…

1 unique partner identity



BETWEEN-PDB HETEROGENEITY

HIGH

median IQR	0.57
90th percentile	1.60
supported positions	205 / 230

CONSTRUCTS & CRYSTALS

3 construct patterns · 5 crystal forms

2.30 Å median (2.11–2.50)

engineered mutants: 20% (top C229S)

ERRβ NR3B2 · 2 usable PDBs

mixed apo/holo · one dominant ligand · partner-rich · IQR not assessable

STRUCTURES

HOLO 1 | APO 1

3 found · 2 with coordinates · 92% of LBD observed (median)

LIGANDS

1 unique · 2OH in 100%

2OH 1

2 ligand–partner contexts

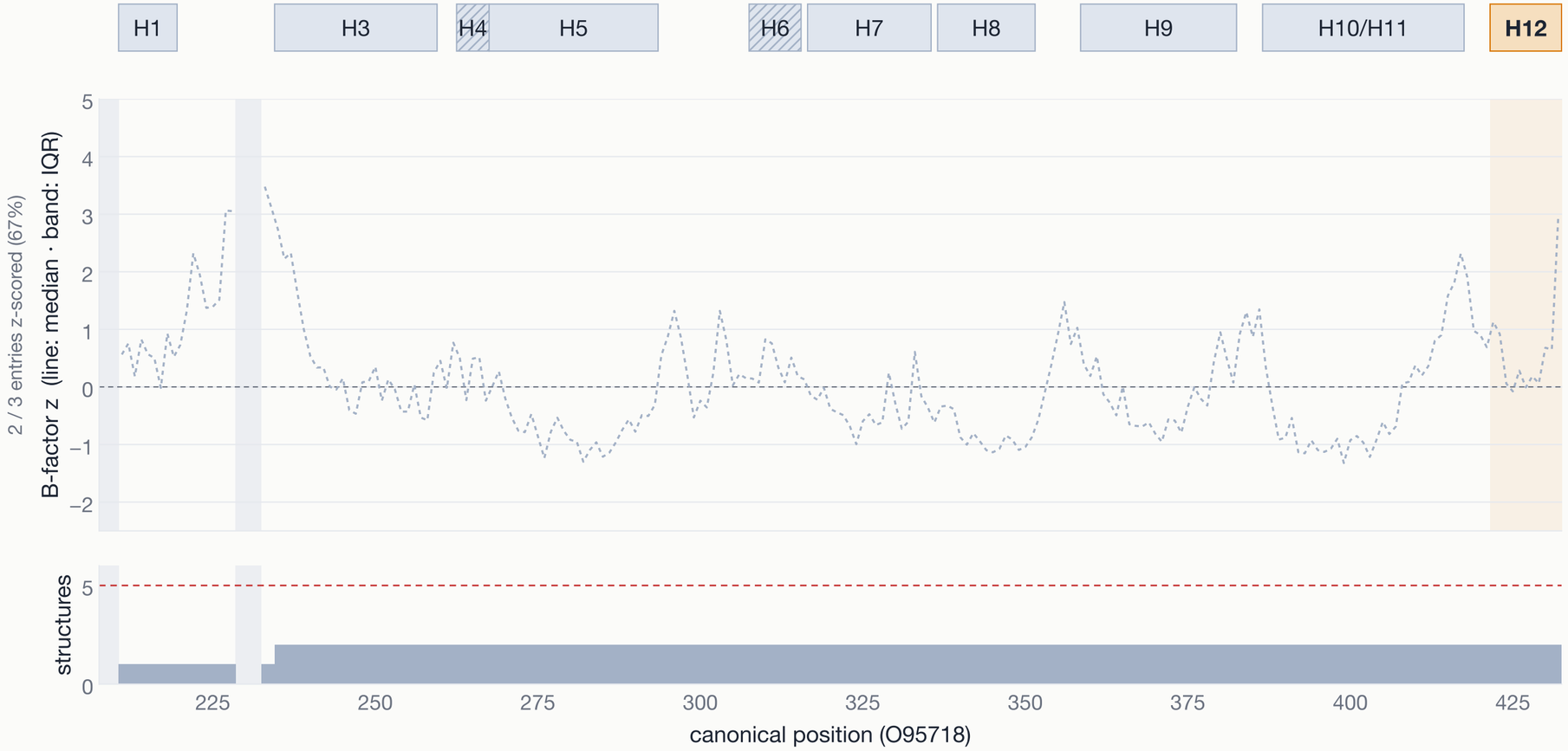
dominant: 2OH + NCOA2 (50%)

PARTNERS / COFACTORS

coregulator 2

most common: 10-mer from nuclear receptor coactivator 2

2 unique partner identities



BETWEEN-PDB HETEROGENEITY

not assessable

Only 2 usable structures

CONSTRUCTS & CRYSTALS

2 construct patterns · 2 crystal forms

2.30 Å median

engineered mutants: 100% (top Y356H)

ERRγ NR3B3 · 33 usable PDBs

mixed apo/holo · ligand-diverse · partly partner-bound · low IQR heterogeneity

STRUCTURES

HOLO 27 | APO 5

TRACE 1

43 found · 33 with coordinates · 100% of LBD observed (median)

LIGANDS

23 unique

OHT 4 · 2OH 3 · TXF 1

26 ligand–partner contexts

dominant: 4-hydroxytamoxifen + standalone (9%)

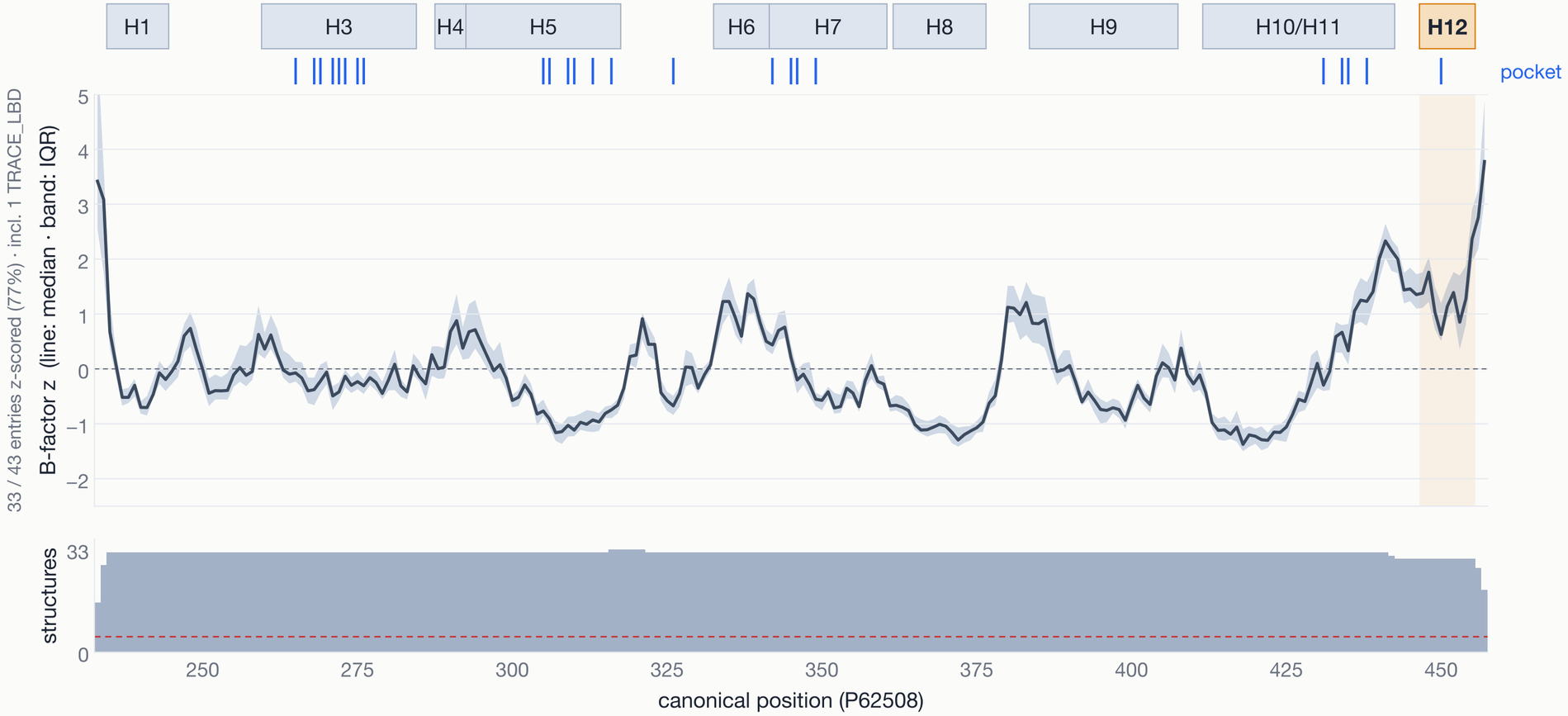
PARTNERS / COFACTORS

other 8 | standalone 22

coregulator 3

most common: nuclear receptor-interacting protein 1

3 unique partner identities



BETWEEN-PDB HETEROGENEITY

LOW

median IQR	0.42
90th percentile	0.74
supported positions	225 / 225

CONSTRUCTS & CRYSTALS

5 construct patterns · 12 crystal forms

1.95 Å median (1.70–2.45)

CAVEATS

- 1 TRACE_LBD fragment (receptor peptides) excluded from apo/holo.

GR NR3C1 · 33 usable PDBs

holo only · ligand-diverse · partner-rich · high IQR heterogeneity

STRUCTURES

HOLO 29 | no APO

TRACE 4

57 found · 33 with coordinates · 98% of LBD observed (median)

LIGANDS

24 unique

DEX 3 · 486 3 · GW6 2

26 ligand–partner contexts

dominant: dexamethasone + NCOA2 (10%)

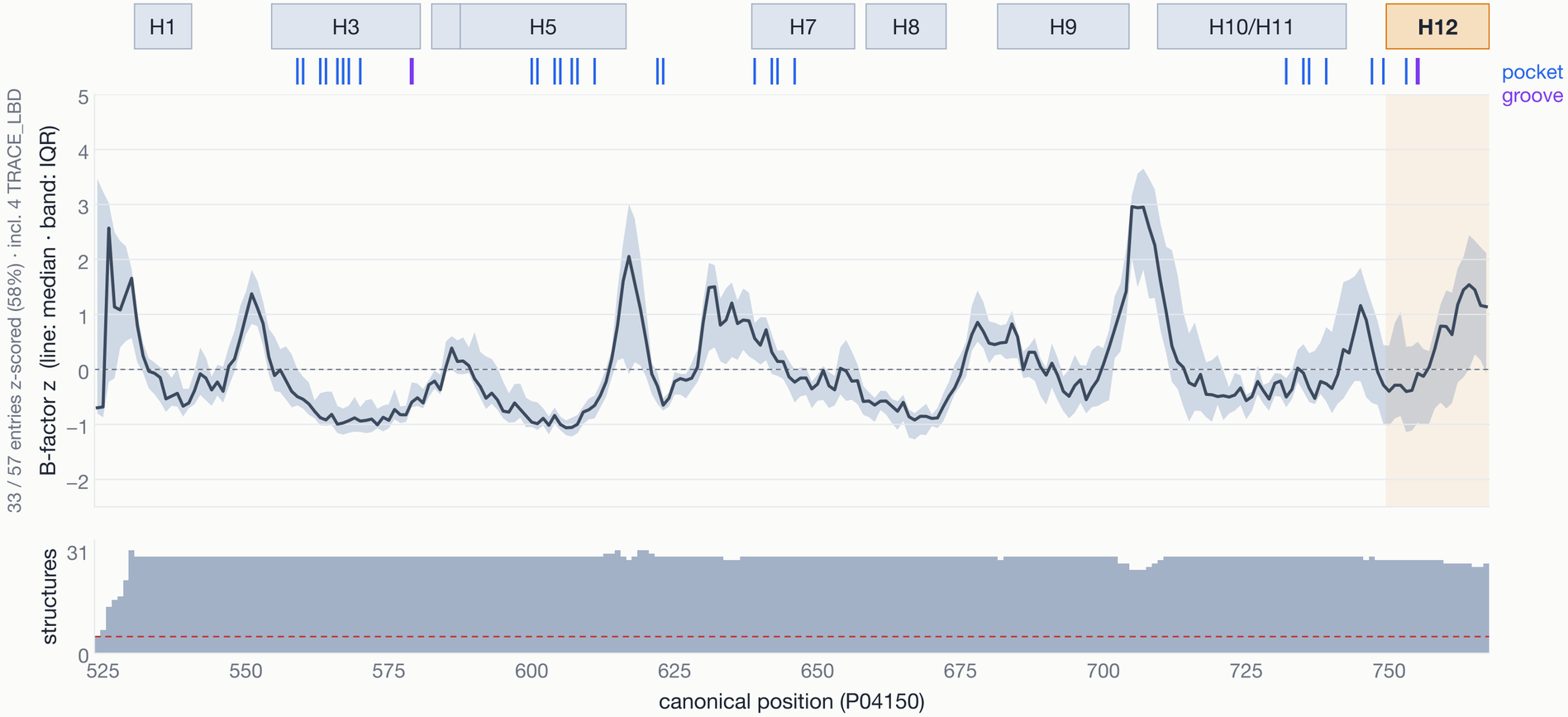
PARTNERS / COFACTORS

coregulator 23 | other 6

standalone 4

most common: nuclear receptor coactivator 2

5 unique partner identities



BETWEEN-PDB HETEROGENEITY

HIGH

median IQR	0.68
90th percentile	1.97
supported positions	244 / 244

CONSTRUCTS & CRYSTALS

12 construct patterns · 23 crystal forms

2.30 Å median (2.13–2.50)

engineered mutants: 82% (top F602S)

CAVEATS

- 4 TRACE_LBD fragments (receptor peptides) excluded from apo/holo.

MR NR3C2 · 29 usable PDBs

holo only · ligand-diverse · partly partner-bound · high IQR heterogeneity

STRUCTURES

HOLO 29 — no APO

30 found · 29 with coordinates · 95% of LBD observed (median)

LIGANDS

24 unique

1CA 3 · STR 3 · SNL 3

23 ligand–partner contexts

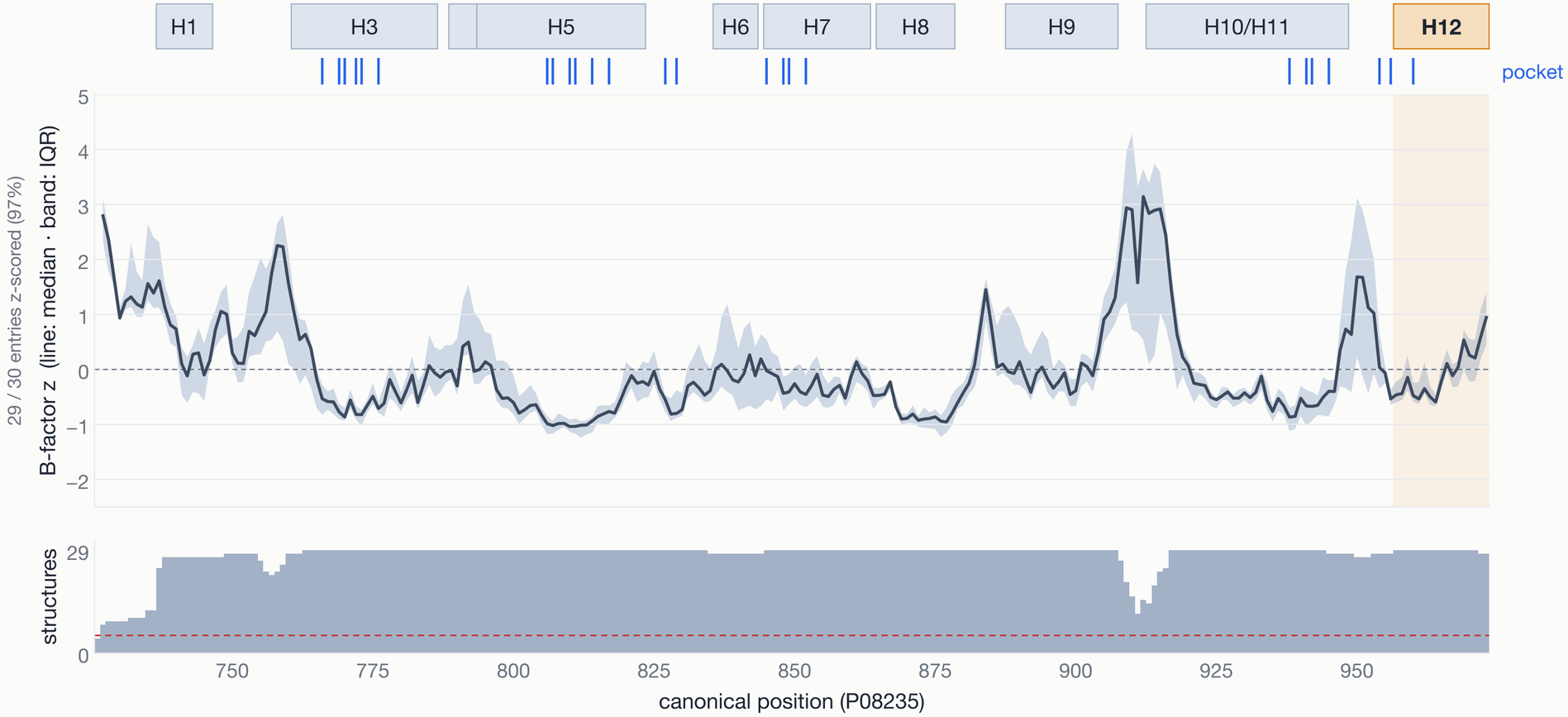
dominant: desoxycorticosterone + standalone (100%)

PARTNERS / COFACTORS

coregulator 12 | standalone 17

most common: nuclear receptor coactivator 1

1 unique partner identity



BETWEEN-PDB HETEROGENEITY

HIGH

median IQR	0.58
90th percentile	1.40
supported positions	247 / 248

CONSTRUCTS & CRYSTALS

5 construct patterns · 15 crystal forms

1.95 Å median (1.75–2.20)

engineered mutants: 76% (top C808S)

substitution status uncertain: 24%

CAVEATS

- Very heterogeneous crystal forms (15 forms in 29 structures).

PR NR3C3 · 19 usable PDBs

holo only · ligand-diverse · partly partner-bound · moderate IQR heterogeneity

STRUCTURES

HOLO 19 — no APO

21 found · 19 with coordinates · 98% of LBD observed (median)

LIGANDS

14 unique

AS0 3 · OR8 3 · NDR 2

16 ligand–partner contexts

dominant: OR8 + standalone (16%)

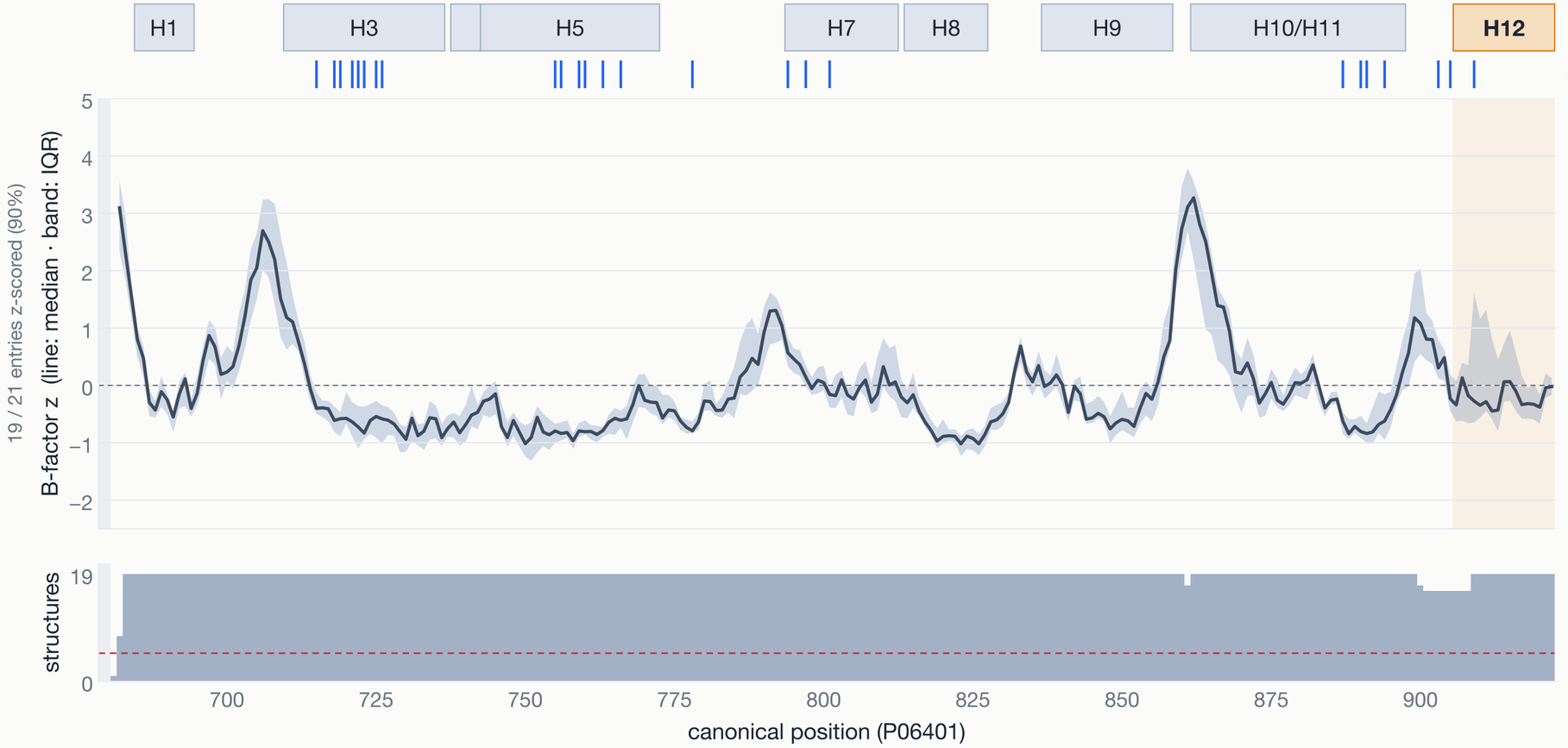
PARTNERS / COFACTORS

standalone 16

coregulator 3

most common: smrt peptide

2 unique partner identities



BETWEEN-PDB HETEROGENEITY

MODERATE

median IQR	0.52
90th percentile	1.17
supported positions	241 / 244

CONSTRUCTS & CRYSTALS

5 construct patterns · 3 crystal forms

2.00 Å median (1.80–2.06)

AR NR3C4 · 92 usable PDBs

holo only · mixed ligands · partly partner-bound · low IQR heterogeneity

STRUCTURES

HOLO 92 — no APO

95 found · 92 with coordinates · 97% of LBD observed (median)

LIGANDS

51 unique

DHT 35 · TES 11 · 198 8

60 ligand–partner contexts

dominant: DHT + standalone (8%)

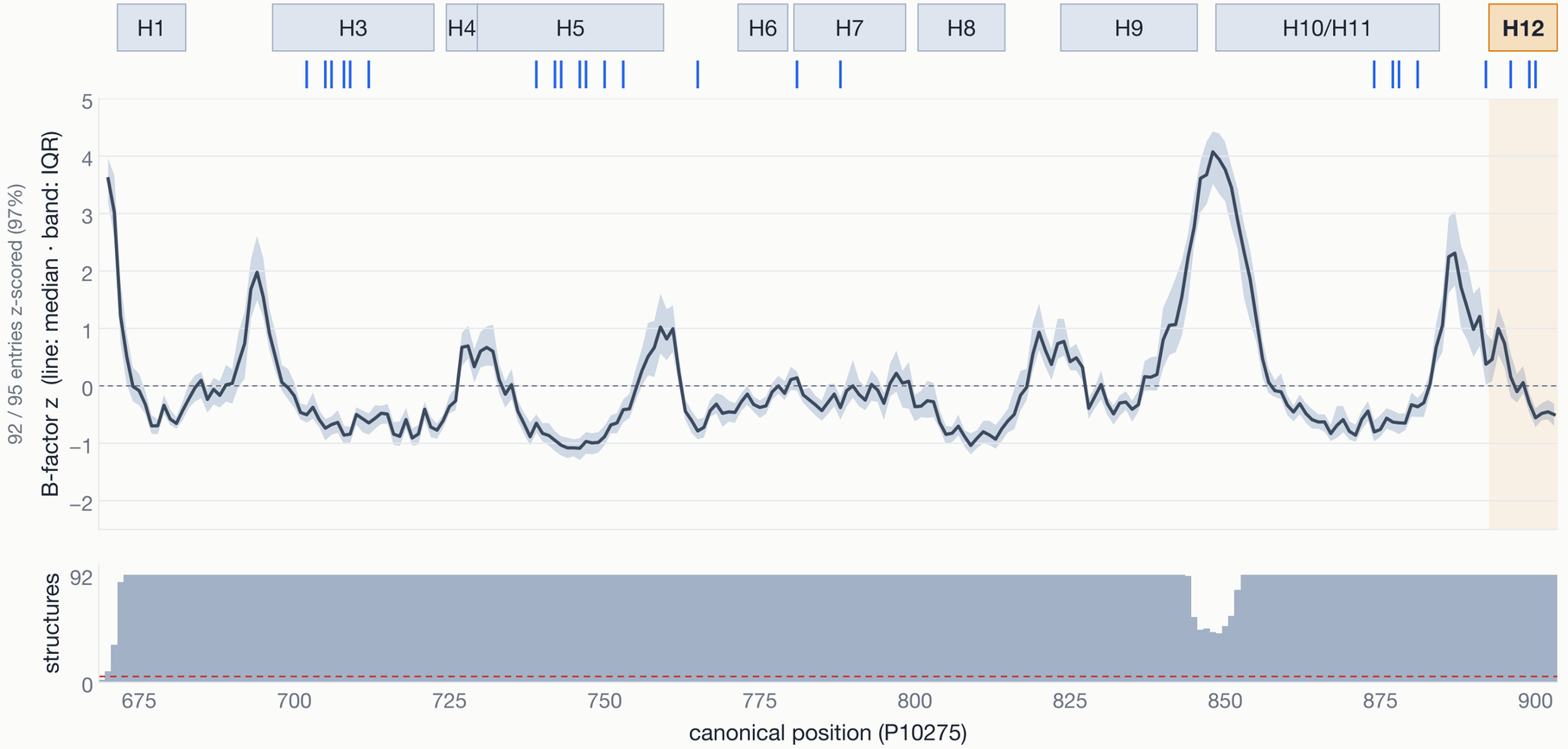
PARTNERS / COFACTORS

other 26 | standalone 57

coregulator 8 NR partner 1

most common: co-regulator peptide

7 unique partner identities



BETWEEN-PDB HETEROGENEITY

LOW

median IQR	0.41
90th percentile	1.03
supported positions	234 / 235

CONSTRUCTS & CRYSTALS

16 construct patterns · 10 crystal forms

1.98 Å median (1.80–2.31)

engineered mutants: 41% (top R761A)

NUR77 NR4A1 · 21 usable PDBs

mixed apo/holo · ligand-diverse · mostly standalone · low IQR heterogeneity

STRUCTURES

HOLO 9 | APO 12

22 found · 21 with coordinates · 98% of LBD observed (median)

LIGANDS

9 unique

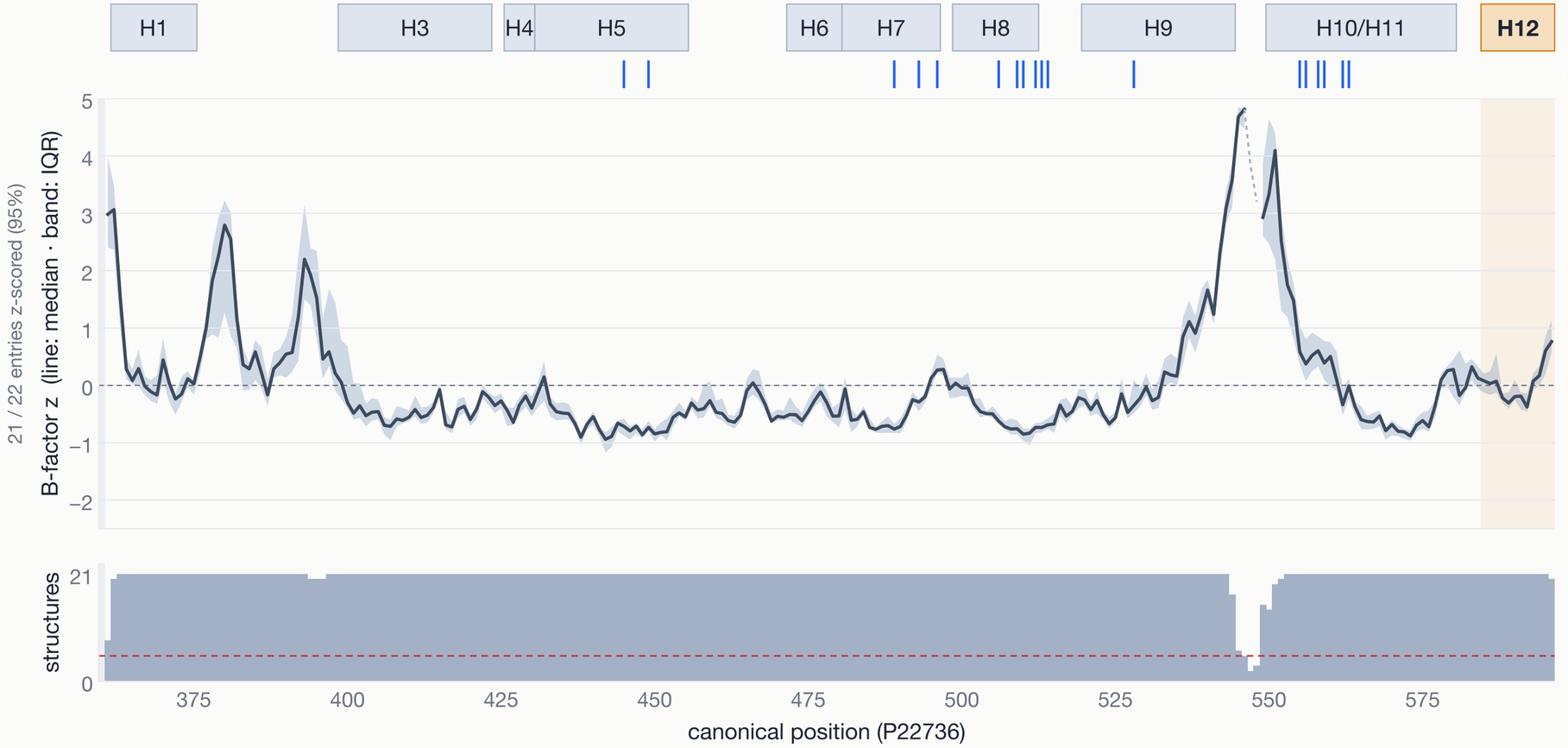
T94 1 · 3MJ 1 · 3MZ 1

10 ligand–partner contexts

dominant: APO + standalone (57%)

PARTNERS / COFACTORS

Standalone: 100%



BETWEEN-PDB HETEROGENEITY

LOW

median IQR	0.33
90th percentile	0.85
supported positions	234 / 237

CONSTRUCTS & CRYSTALS

3 construct patterns · 2 crystal forms

2.37 Å median (2.18–2.60)

engineered mutants: 24% (top L449W)

NURR1 NR4A2 · 6 usable PDBs

apo only · no bound ligands · partner-rich · moderate IQR heterogeneity

STRUCTURES

APO 6

8 found · 6 with coordinates · 97% of LBD observed (median)

LIGANDS

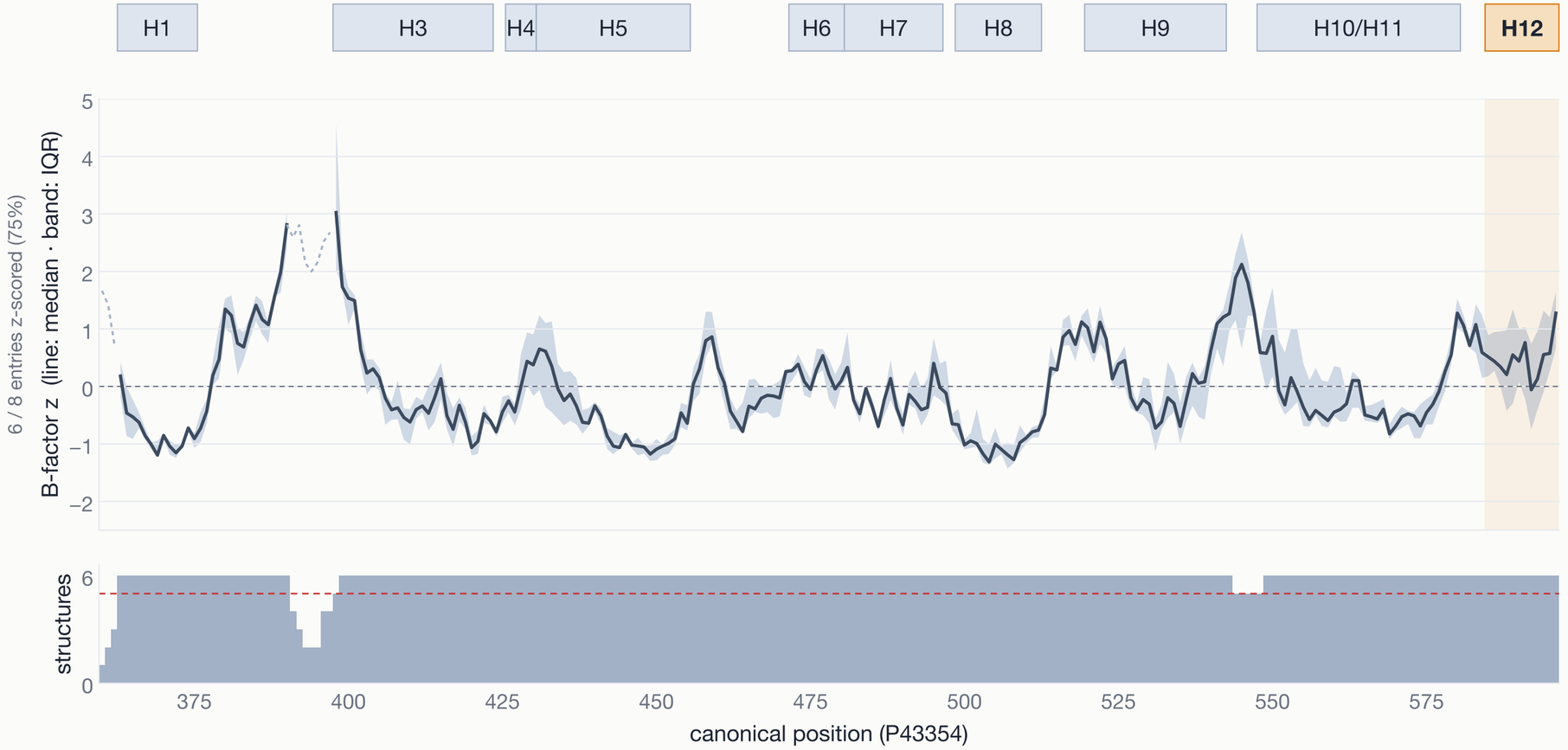
no bound ligands

1 ligand–partner context
dominant: APO + standalone (100%)

PARTNERS / COFACTORS

other 3 | standalone 3

most common: nuclear receptor subfamily 4 group a…



BETWEEN-PDB HETEROGENEITY

MODERATE

median IQR	0.50
90th percentile	0.98
supported positions	227 / 237

CONSTRUCTS & CRYSTALS

3 construct patterns · 6 crystal forms
2.38 Å median (2.24–2.93)

NOR-1 NR4A3 · 0 usable PDBs

no usable LBD structures

STRUCTURES

No usable LBD structures

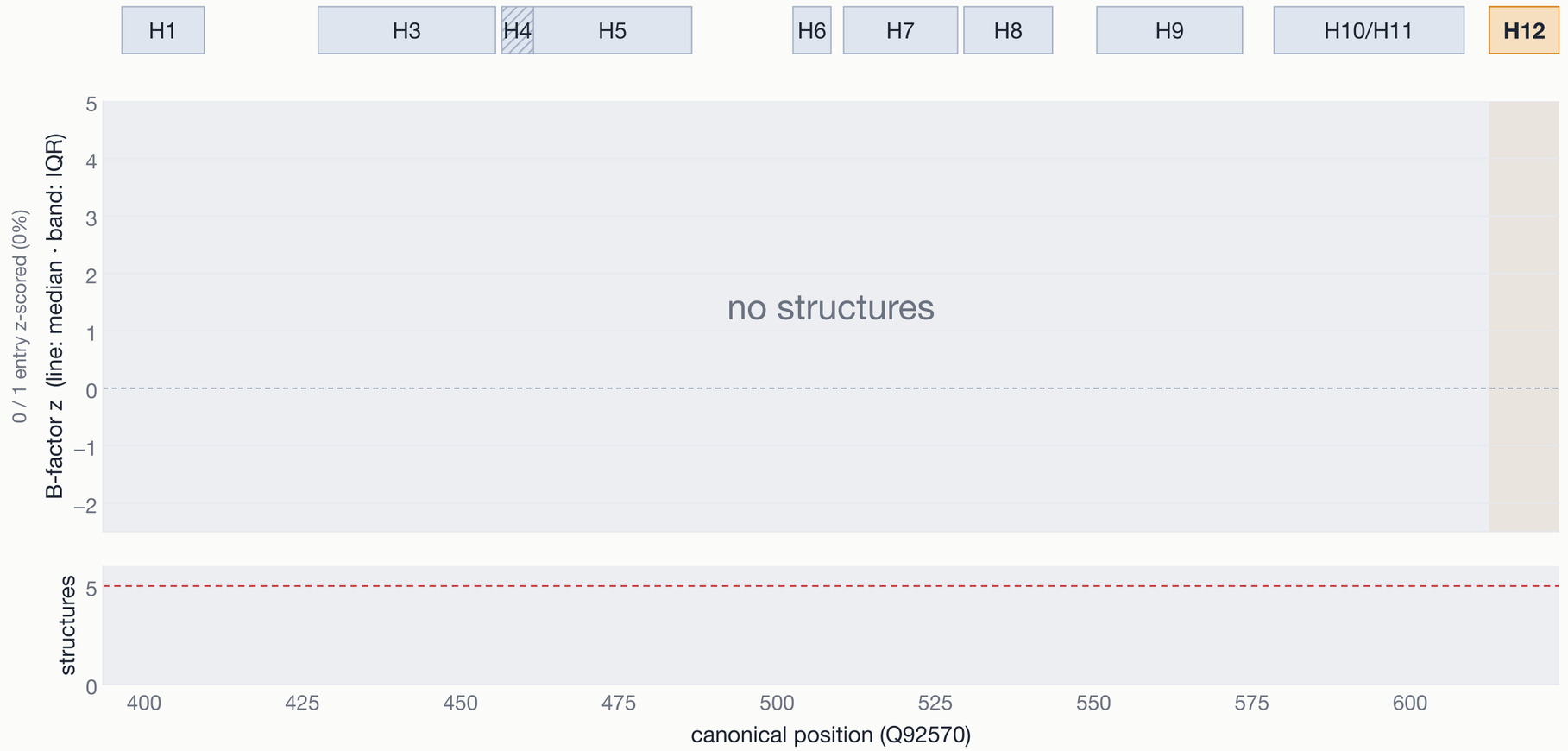
1 entry found (not X-ray (NMR/EM) (1))

LIGANDS

–

PARTNERS / COFACTORS

–



BETWEEN-PDB HETEROGENEITY

not assessable

No structures

SF-1 NR5A1 · 6 usable PDBs

holo only · ligand-diverse · partner-rich · low IQR heterogeneity

STRUCTURES

HOLO 6 no APO

6 found · 6 with coordinates · 99% of LBD observed (median)

LIGANDS

6 unique

PEF 2 · P0E 1 · PIZ 1

6 ligand–partner contexts

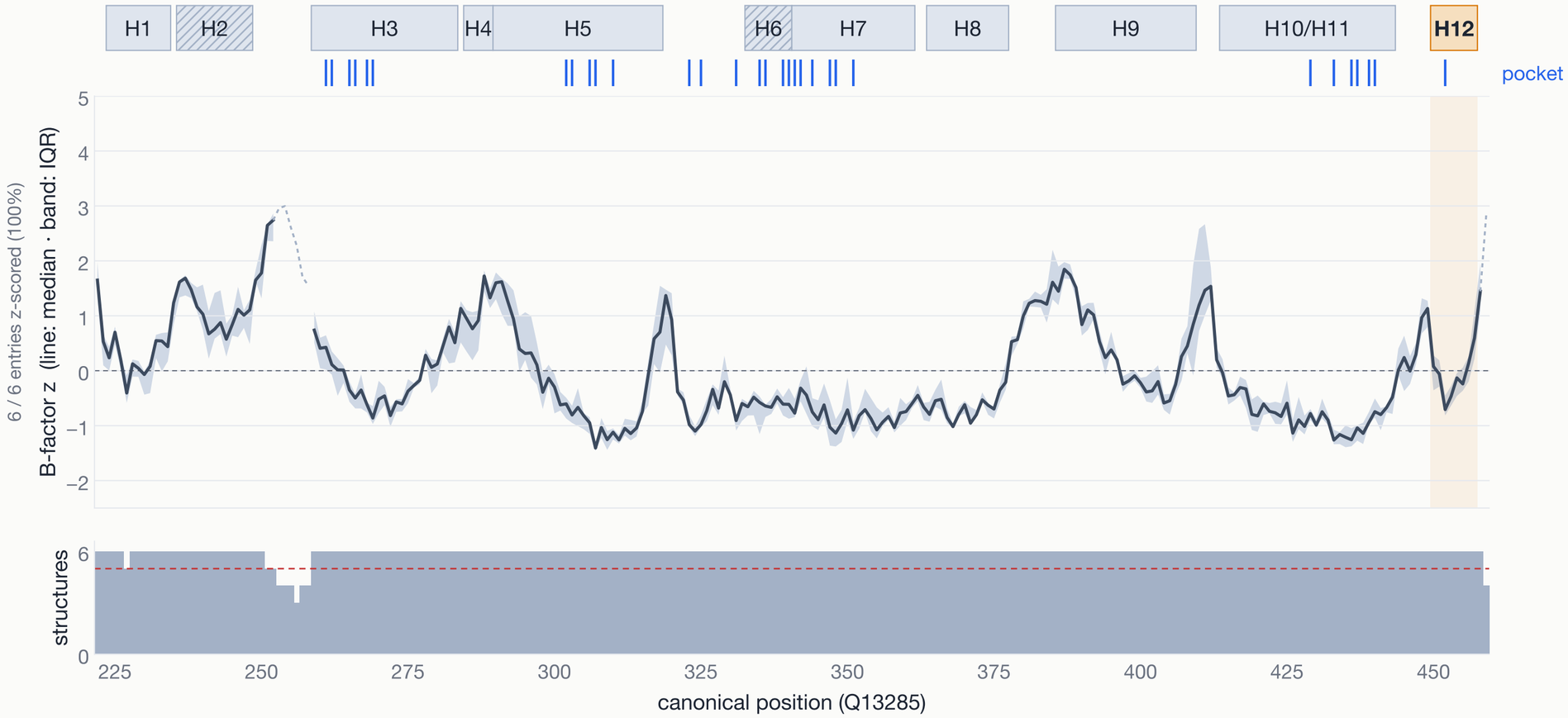
dominant: phosphatidyl ethanol + NCOA2 (17%)

PARTNERS / COFACTORS

coregulator 6

most common: peroxisome proliferator-activated…

2 unique partner identities



BETWEEN-PDB HETEROGENEITY

LOW

median IQR	0.38
90th percentile	0.81
supported positions	231 / 238

CONSTRUCTS & CRYSTALS

3 construct patterns · 4 crystal forms

2.55 Å median (2.43–2.76)

engineered mutants: 67% (top C247S)

1 internal deletion

LRH-1 NR5A2 · 25 usable PDBs

mostly holo · ligand-diverse · partner-rich · high IQR heterogeneity

STRUCTURES

HOLO 23

APO 2

28 found · 25 with coordinates · 99% of LBD observed (median)

LIGANDS

18 unique

EPH 4 · P6L 2 · RJW 2

23 ligand–partner contexts

dominant: EPH + SHP (12%)

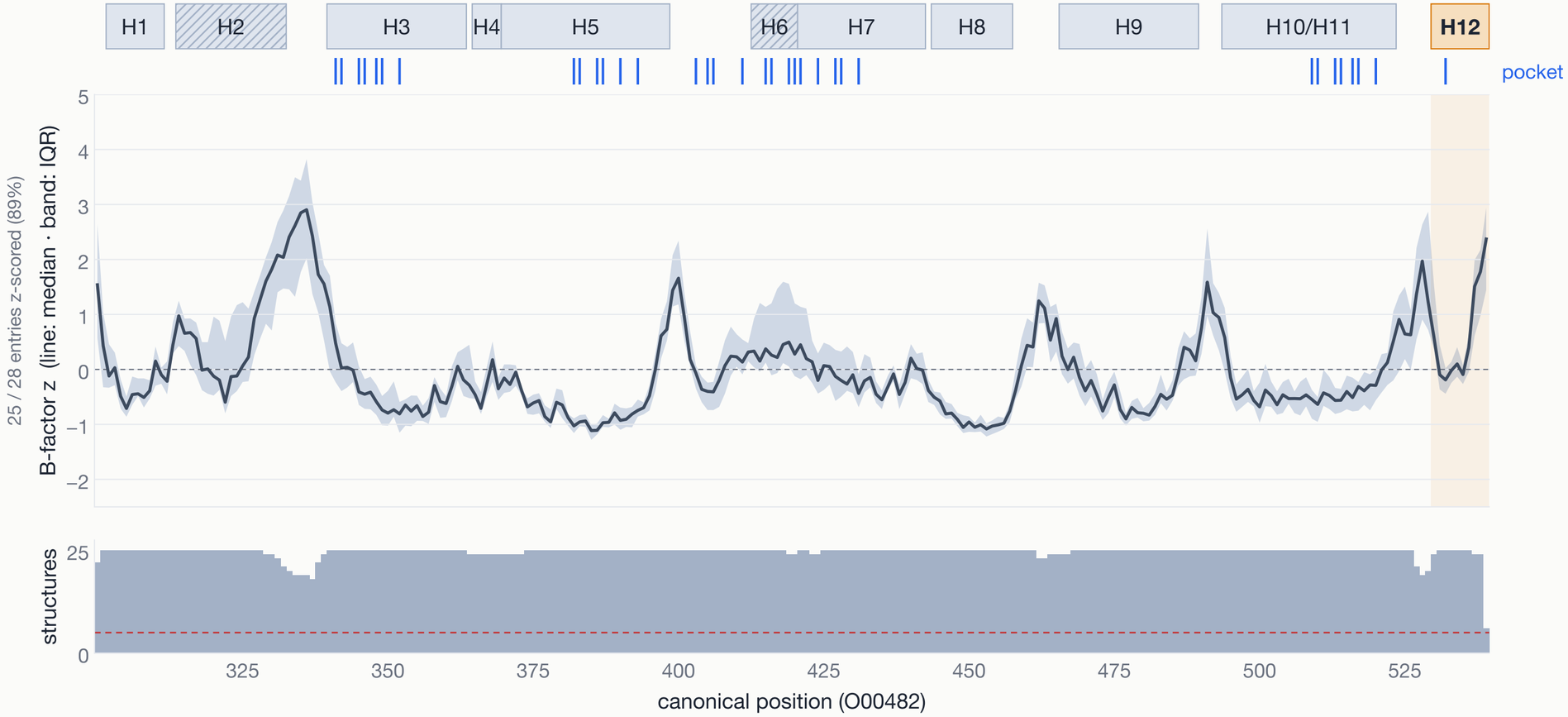
PARTNERS / COFACTORS

coregulator 19

NR partner 4 other 1 standalone 1

most common: nuclear receptor coactivator 2

5 unique partner identities



BETWEEN-PDB HETEROGENEITY

HIGH

median IQR	0.60
90th percentile	1.37
supported positions	240 / 240

CONSTRUCTS & CRYSTALS

11 construct patterns · 14 crystal forms

2.00 Å median (1.86–2.50)

CAVEATS

- Very heterogeneous crystal forms (14 forms in 25 structures).

GCNF NR6A1 · 0 usable PDBs

no usable LBD structures

STRUCTURES

No usable LBD structures

0 entries found

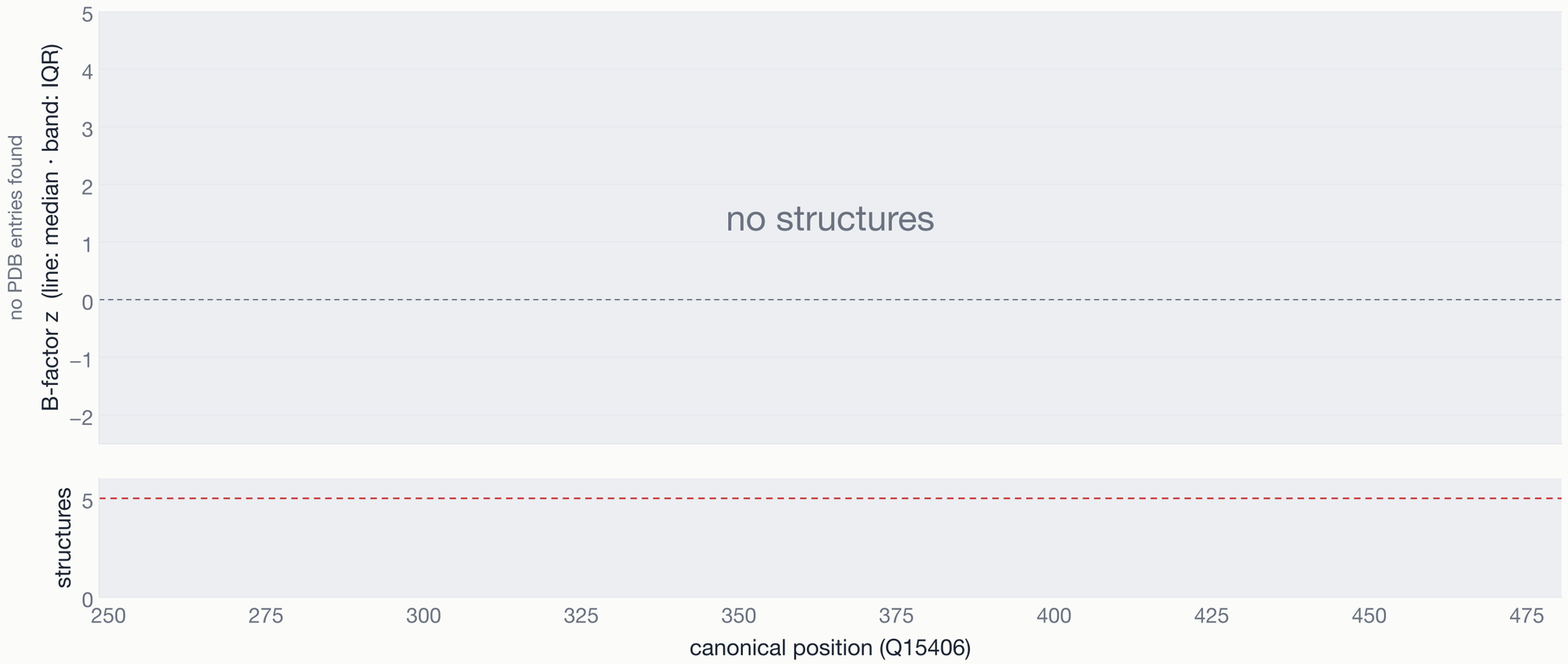
LIGANDS

–

PARTNERS / COFACTORS

–

no professor helix map



BETWEEN-PDB HETEROGENEITY

not assessable

No structures

CAVEATS

- No professor helix map: no canonical elements.